MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

GATE, NAND, DIGITAL, KEY CODE RBF

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic NAND gate module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

* 2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.534 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be RBF.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/83-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 V dc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

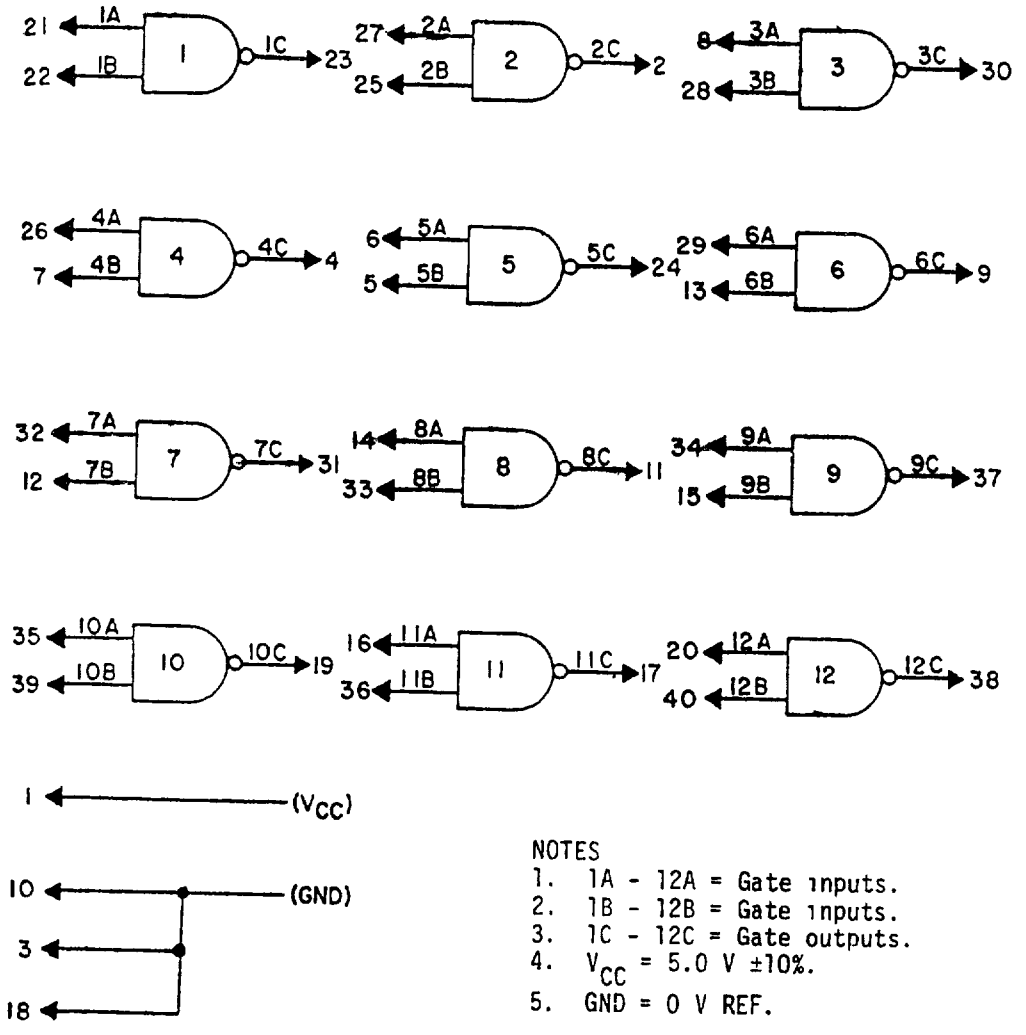
4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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NOTES
1. 1A - 12A = Gate inputs.
2. 1B - 12B = Gate inputs.
3. 1C - 12C = Gate outputs.
4. V_{CC} = 5.0 V ±10%.
5. GND = 0 V REF.

FIGURE 1. RBF module function diagram, NAND gate.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation	III	S	10	∞	1	∞	10	∞	1	∞	MΩ
B1-B12	V_{OUT} (low loaded)	IV	S	0.01	1.00	0.01	1.00	0.01	1.00	0.01	1.00	V
C1-C24	Input breakdown current	V	S	0	1	0	1	0	1	0	1	mA
D1-D12	Output leakage current	VI	X	0	1	0	1	0	1	0	1	mA
E1-E12	Output short circuit current	VII	X	-40	-100	-40	-100	-40	-100	-40	-100	mA
F1-F24	Input clamp diode voltage	VIII	X	-0.4	-1.2	-0.4	-1.2	-0.4	-1.2	-0.4	-1.2	V
G1-G12	Output voltage, low level	IX	X	0	500	0	500	0	500	0	500	mV
H1-H24	Output voltage, high level	X	X	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	V
J1-J24	Input current, high level	XI	X	0	50	0	50	0	50	0	50	µA
K1-K24	Input current, low level	XII	X	-0.4	-2.2	-0.4	-2.2	-0.4	-2.2	-0.4	-2.2	mA
L1,L2	Ground pin continuity	XIII	X	0	1	0	1	0	1	0	1	Ω
L3,L4	Ground pin continuity	XIII	Q	0.00	0.10	0.00	0.10	0.00	0.10	0.00	0.10	Ω
M1	Power supply current, low level	XIV	S	30	108	30	108	30	108	30	108	mA
M2	Power supply current, high level	XIV	S	15	48	15	48	15	48	15	48	mA
N1-N12	Transition time (t_{TLH}), C_L = 50 pF	XV	S	2	15	2	15	2	18	2	18	ns
N13-N24	Transition time (t_{TLH}), C_L = 150 pF	XV	Q	2	27	2	27	2	30	2	30	ns
P1-P12	Transition time (t_{THL}), C_L = 50 pF	XV	S	2	6	2	6	2	8	2	8	ns
P13-P24	Transition time (t_{THL}), C_L = 150 pF	XV	Q	2	11	2	11	2	13	2	13	ns

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
Q1-Q12	Propagation delay time (t_{PLH}), C_L = 50 pF	XV	S	2	7	2	7	2	8	2	8	ns
Q13-Q24	Propagation delay time (t_{PLH}), C_L = 150 pF	XV	Q	4.0	10.5	4.0	10.5	4	12	4	12	ns
R1-R12	Propagation delay time (t_{PHL}), C_L = 50 pF	XV	S	2.0	7.5	2.0	7.5	2	8	2	8	ns
R13-R24	Propagation delay time (t_{PHL}), C_L = 150 pF	XV	Q	4.0	11.5	4.0	11.5	4	12	4	12	ns
S1-S12	Output to output delta (Δt_{PLH}), C_L = 50 pF	XVI	S	-3	3	-3	3	-3	3	-3	3	ns
T1-T12	Output to output delta propagation delay time (Δt_{PHL}), C_L = 50 pF	XVI	S	-3	3	-3	3	-3	3	-3	3	ns
U1-U12	Output crosstalk, low level	XVII	Q	0	500	0	500	0	500	0	500	mV
V1-V12	Output crosstalk, high level	XVII	Q	0	400	0	400	0	400	0	400	mV
W	Power supply transient amplitude	XVIII	Q	0	600	0	600	0	600	0	600	mV
X	Power supply filter capacitance	XIX	S	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	µF
Y1-Y24	Input capacitance	XX	Q	2	15	2	15	2	15	2	15	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q' parameters shall be tested during qualification testing only.

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4.4.2 Order of testing. Test conditions A through Y shall be performed in any order with the following exceptions. If test conditions A, B, C, D, or E are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D, E where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through Y in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through X shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition X, shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through X shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through Y shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Cross reference of classification. Part number M28787/83-1 supersedes part number M28787/83. The two parts are interchangeable.

6.4 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life limits from table I.

Supply voltage	5 V dc ±10 percent
Maximum power dissipation	
All outputs at low level	594 mW (outputs not loaded)
All outputs at high level	264 mW (outputs not loaded)
Input voltage	
Low level	-0.5 V minimum, 0.8 V maximum
High level	2.0 V minimum, V_{CC} maximum

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Input current		
Low level	(V_{IL} = 0.0 V)	-2.2 mA maximum
	(V_{IL} = 0.5 V)	-2.0 mA maximum
High level	(V_{IH} = 2.5 V)	50 µA maximum
	(V_{IH} = 5.5 V)	1.0 mA maximum
Output voltage		
Low level	(I_{OL} = 20 mA)	0.5 V maximum
High level	(I_{OH} = 2.5 V)	2.5 V minimum
Output drive current maximum capability		
Low level	(V_{OL} = 0.5 V)	20 mA
High level	(V_{OH} = 2.5 V)	-1.0 mA

6.5 Module function. This module contains 12 independent 2-input NAND gates.

6.5.1 Input to output relationship. Each gate has the input to output relationship of table II for positive logic.

TABLE II. Input to output relationship.

Inputs		Output
A	B	C
L	L	H
L	H	H
H	L	H
H	H	L

NOTES:

1. H = High level.
2. L = Low level.

6.5.2 Parallel outputs. Paralleling of outputs to obtain extended drive is permissible only when parallel inputs and outputs are confined within each group specified below. The output drive capability of this combination is equal to the sum of the output drive capabilities of the individual circuits. The circuit groups which can be paralleled are:

a. Circuits 1, 2, and 3.
b. Circuits 4, 5, and 6.
c. Circuits 7, 8, and 9.
d. Circuits 10, 11, and 12.

6.5.3 Ground pins. All electrical parameters are measured with pins 3, 10, and 18 connected separately to 0 V return. System problems may result from a different grounding method.

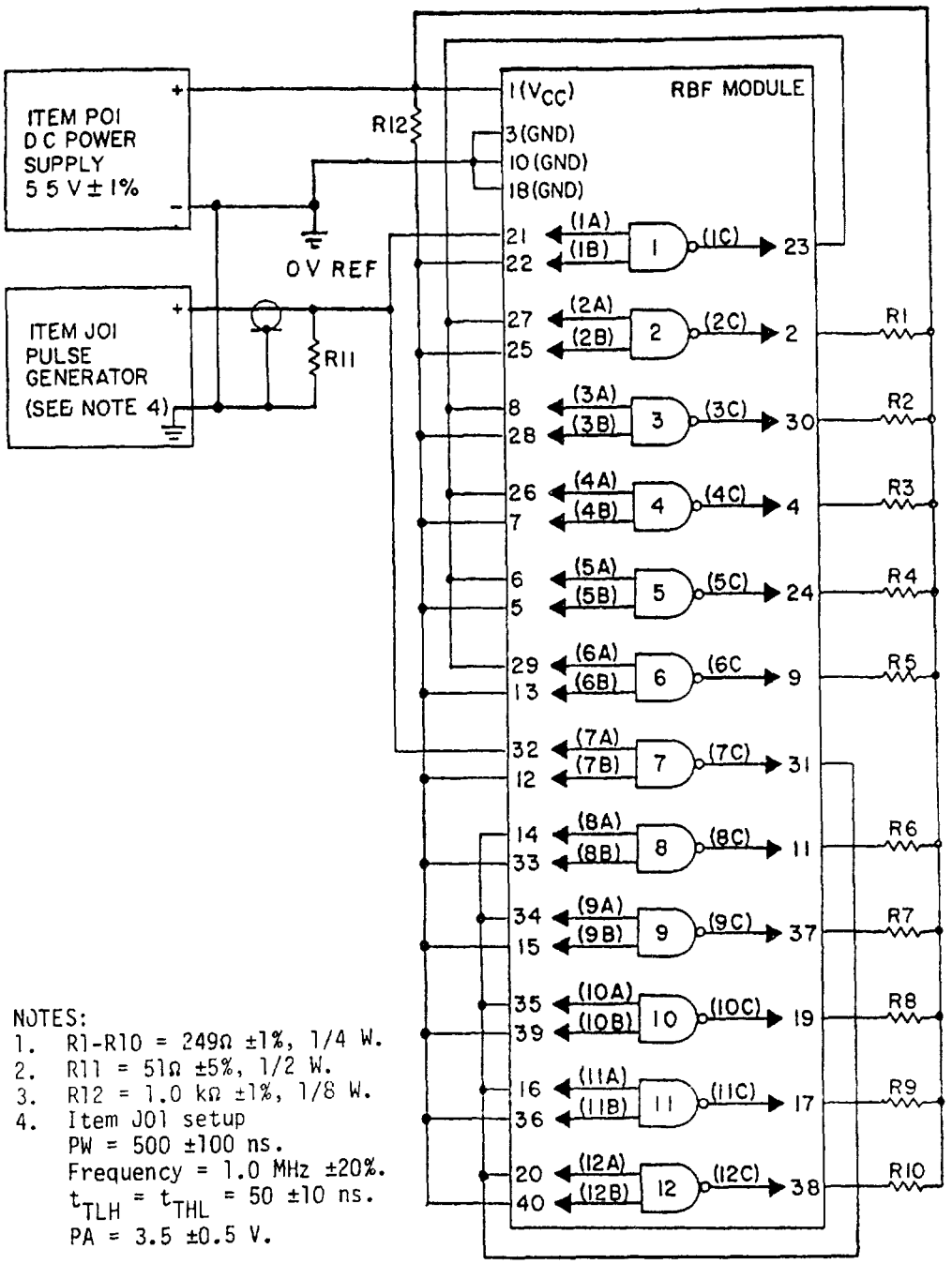
6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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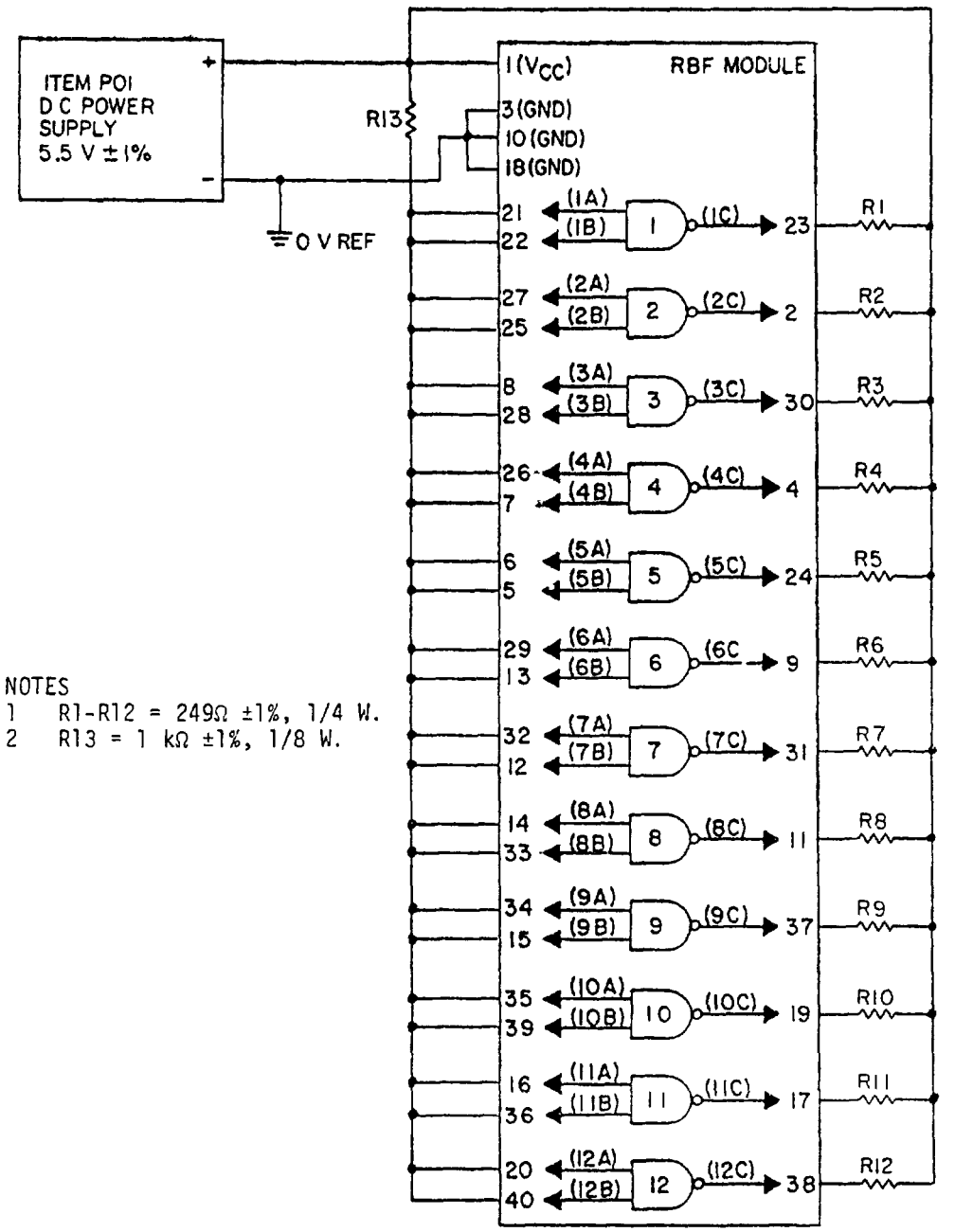
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NOTES:
1. R1-R10 = 249Ω ±1%, 1/4 W.
2. R11 = 51Ω ±5%, 1/2 W.
3. R12 = 1.0 kΩ ±1%, 1/8 W.
4. Item JO1 setup
   PW = 500 ±100 ns.
   Frequency = 1.0 MHz ±20%.
   $t_{TLH} = t_{THL}$ = 50 ±10 ns.
   PA = 3.5 ±0.5 V.

* FIGURE 2. Operating temperature and life test circuit

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NOTES
1 R1-R12 = 249Ω ±1%, 1/4 W.
2 R13 = 1 kΩ ±1%, 1/8 W.

FIGURE 3. Component temperature test circuit

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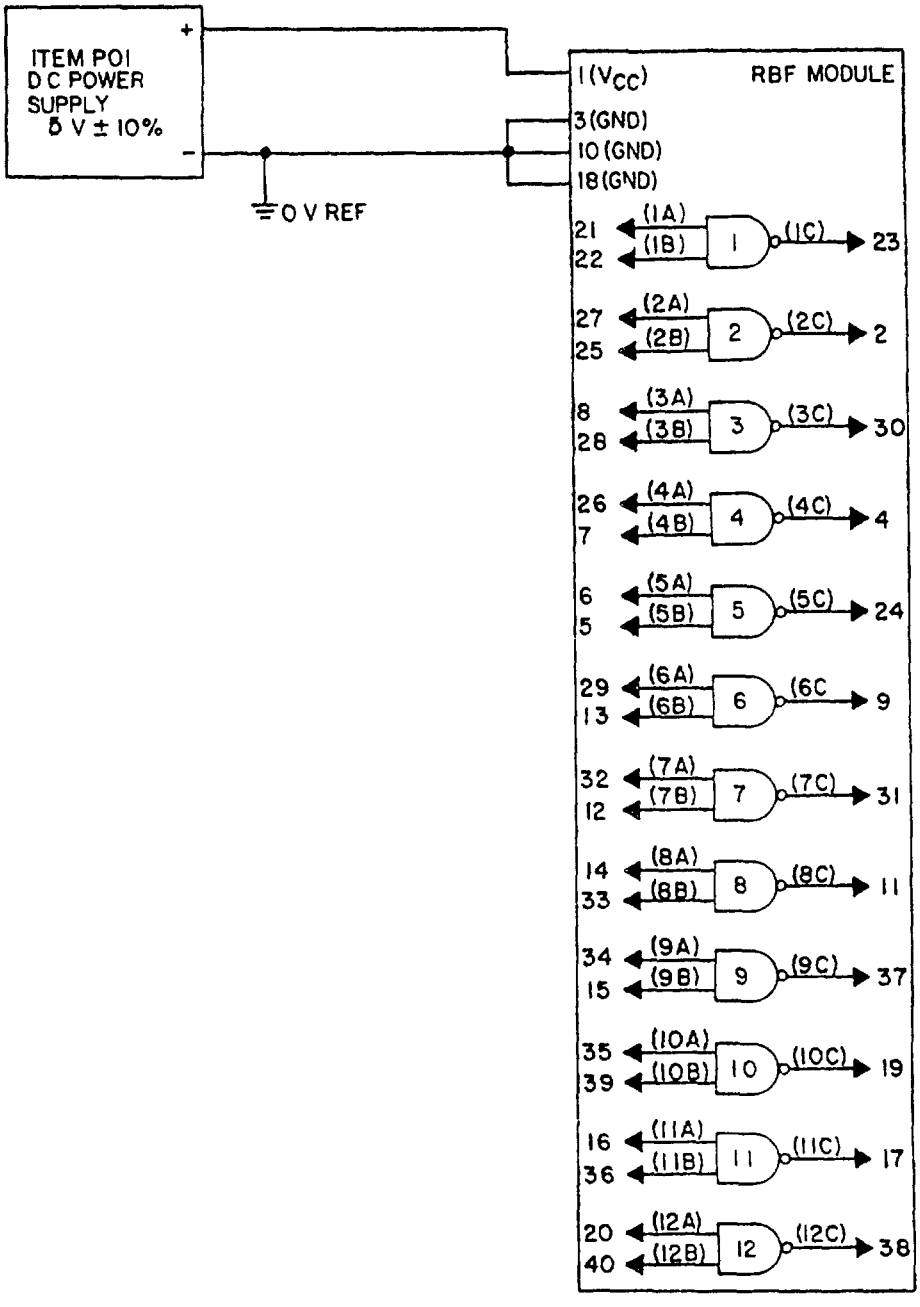


FIGURE 4. Vibration test circuit

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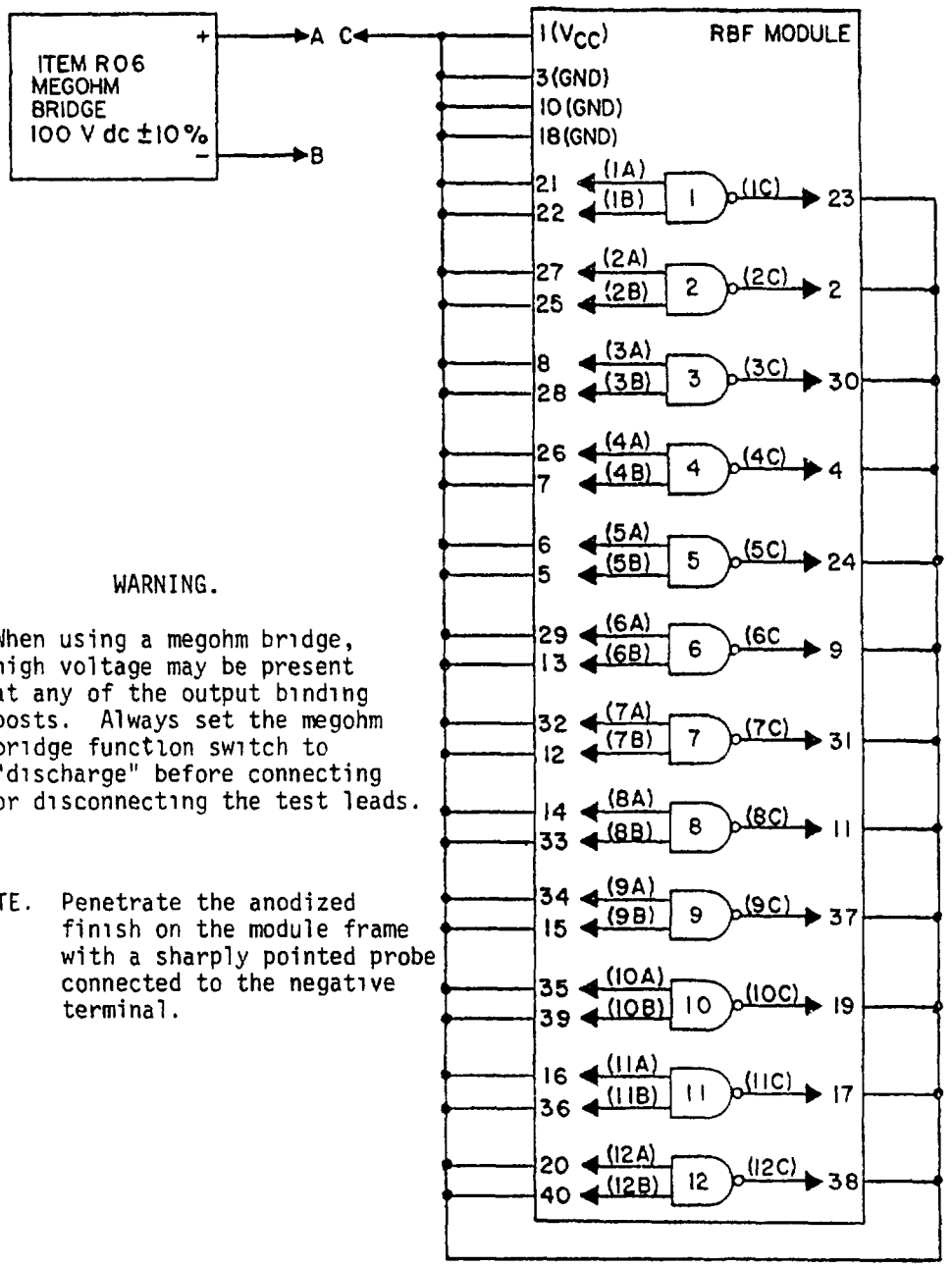
TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A	Isolation	C	Frame	10	∞	MΩ	5

TABLE IV. V_{OUT} (low-loaded) test.

Test condition	Parameter		Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	Min	Max		
B1	1C	V_{OUT} (low loaded)	23	0.01	1.00	V	6
B2	2C	"	2	"	"	"	"
B3	3C	"	30	"	"	"	"
B4	4C	"	4	"	"	"	"
B5	5C	"	24	"	"	"	"
B6	6C	"	9	"	"	"	"
B7	7C	"	31	"	"	"	"
B8	8C	"	11	"	"	"	"
B9	9C	"	37	"	"	"	"
B10	10C	"	19	"	"	"	"
B11	11C	"	17	"	"	"	"
B12	12C	"	38	"	"	"	"

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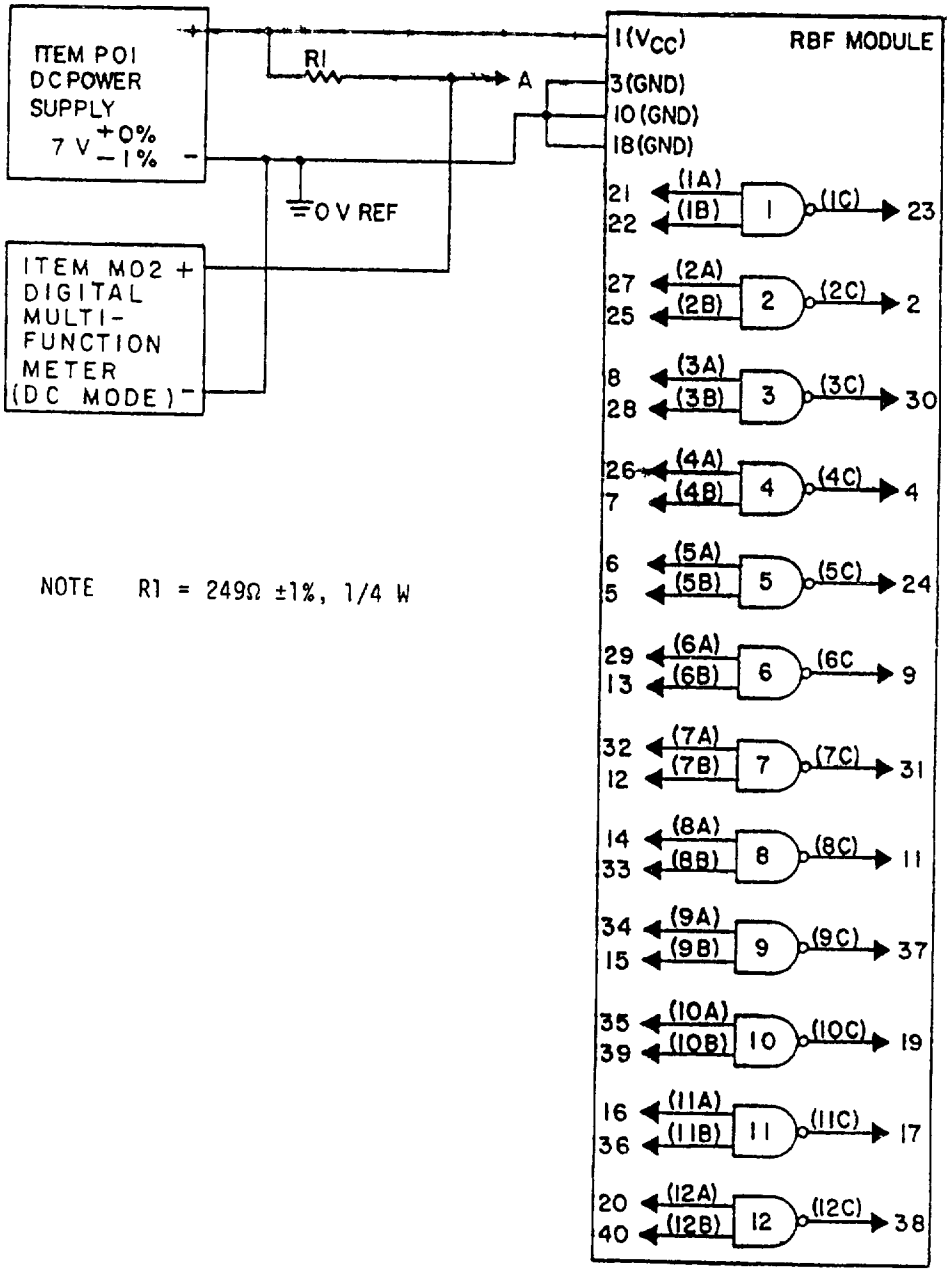
WARNING.

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE. Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 5 Isolation test circuit for table III

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NOTE R1 = 249Ω ±1%, 1/4 W

* FIGURE 6 V_{OUT} (low-loaded) test circuit for table IV

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TABLE V. Input breakdown current test.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
C1	1A	Input breakdown current	5,6,7,8,12,13,14,15,16,20, 22,25,26,27,28,29,32,33,34, 35,36,39,40	21	0	1	mA	7
C2	1B	"	5,6,7,8,12,13,14,15,16,20, 21,25,26,27,28,29,32,33,34, 35,36,39,40	22	"	"	"	"
C3	2A	"	5,6,7,8,12,13,14,15,16,20, 21,22,25,26,28,29,32,33,34, 35,36,39,40	27	"	"	"	"
C4	2B	"	5,6,7,8,12,13,14,15,16,20, 21,22,26,27,28,29,32,33,34, 35,36,39,40	25	"	"	"	"
C5	3A	"	5,6,7,12,13,14,15,16,20,21, 22,25,26,27,28,29,32,33,34, 35,36,39,40	8	"	"	"	"
C6	3B	"	5,6,7,8,12,13,14,15,16,20, 21,22,25,26,27,29,32,33,34, 35,36,39,40	28	"	"	"	"
C7	4A	"	5,6,7,8,12,13,14,15,16,20, 21,22,25,27,28,29,32,33,34, 35,36,39,40	26	"	"	"	"
C8	4B	"	5,6,8,12,13,14,15,16,20,21, 22,25,26,27,28,29,32,33,34, 35,36,39,40	7	"	"	"	"
C9	5A	"	5,7,8,12,13,14,15,16,20,21, 22,25,26,27,28,29,32,33,34, 35,36,39,40	6	"	"	"	"
C10	5B	"	6,7,8,12,13,14,15,16,20,21, 22,25,26,27,28,29,32,33,34, 35,36,39,40	5	"	"	"	"

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TABLE V. Input breakdown current test - Continued.

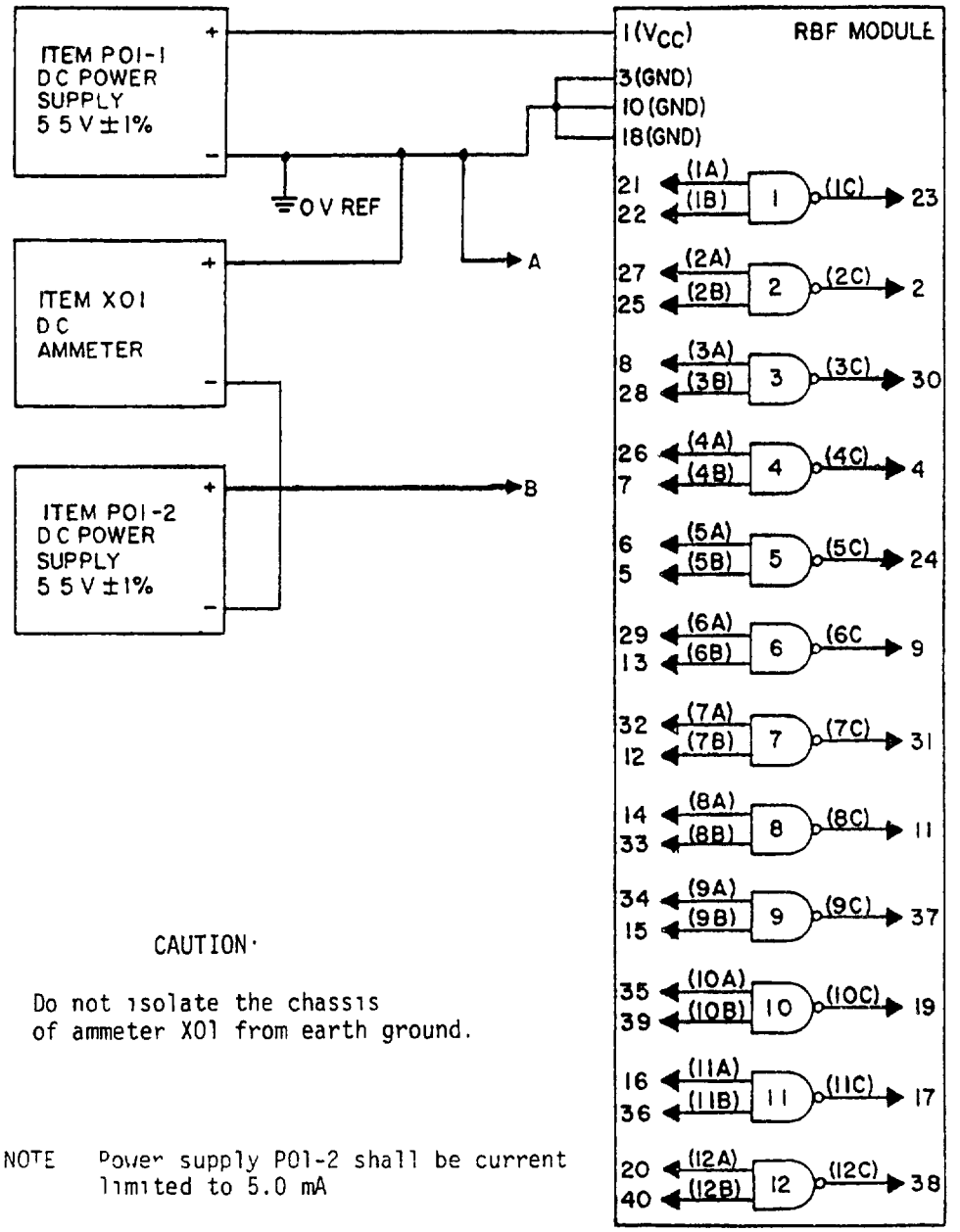
Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
C11	6A	Input breakdown current	5,6,7,8,12,13,14,15,16,20, 21,22,25,26,27,28,32,33,34, 35,36,39,40	29	0	1	mA	7
C12	6B	"	5,6,7,8,12,14,15,16,20,21, 22,25,26,27,28,29,32,33,34, 35,36,39,40	13	"	"	"	"
C13	7A	"	5,6,7,8,12,13,14,15,16,20, 21,22,25,26,27,28,29,33,34, 35,36,39,40	32	"	"	"	"
C14	7B	"	5,6,7,8,13,14,15,16,20,21, 22,25,26,27,28,29,32,33,34, 35,36,39,40	12	"	"	"	"
C15	8A	"	5,6,7,8,12,13,15,16,20,21, 22,25,26,27,28,29,32,33,34, 35,36,39,40	14	"	"	"	"
C16	8B	"	5,6,7,8,12,13,14,15,16,20, 21,22,25,26,27,28,29,32,34, 35,36,39,40	33	"	"	"	"
C17	9A	"	5,6,7,8,12,13,14,15,16,20, 21,22,25,26,27,28,29,32,33, 35,36,39,40	34	"	"	"	"
C18	9B	"	5,6,7,8,12,13,14,16,20,21, 22,25,26,27,28,29,32,33,34, 35,36,39,40	15	"	"	"	"
C19	10A	"	5,6,7,8,12,13,14,15,16,20, 21,22,25,26,27,28,29,32,33, 34,36,39,40	35	"	"	"	"
C20	10B	"	5,6,7,8,12,13,14,15,16,20, 21,22,25,26,27,28,29,32,33, 34,35,36,40	39	"	"	"	"

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TABLE V. Input breakdown current test - Continued.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
C21	11A	Input breakdown current	5,6,7,8,12,13,14,15,20,21, 22,25,26,27,28,29,32,33,34, 35,36,39,40	16	0	1	mA	7
C22	11B	"	5,6,7,8,12,13,14,15,16,20, 21,22,25,26,27,28,29,32,33, 34,35,39,40	36	"	"	"	"
C23	12A	"	5,6,7,8,12,13,14,15,16,21, 22,25,26,27,28,29,32,33,34, 35,36,39,40	20	"	"	"	"
C24	12B	"	5,6,7,8,12,13,14,15,16,20, 21,22,25,26,27,28,29,32,33, 34,35,36,39	40	"	"	"	"

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CAUTION:

Do not isolate the chassis
of ammeter XO1 from earth ground.

NOTE Power supply PO1-2 shall be current limited to 5.0 mA

FIGURE 7. Input breakdown current test circuit for table V.

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TABLE VI. Output leakage current test.

Test condition	Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
D1	1C Output leakage current	23	0	1	mA	8
D2	2C "	2	"	"	"	"
D3	3C "	30	"	"	"	"
D4	4C "	4	"	"	"	"
D5	5C "	24	"	"	"	"
D6	6C "	9	"	"	"	"
D7	7C "	31	"	"	"	"
D8	8C "	11	"	"	"	"
D9	9C "	37	"	"	"	"
D10	10C "	19	"	"	"	"
D11	11C "	17	"	"	"	"
D12	12C "	38	"	"	"	"

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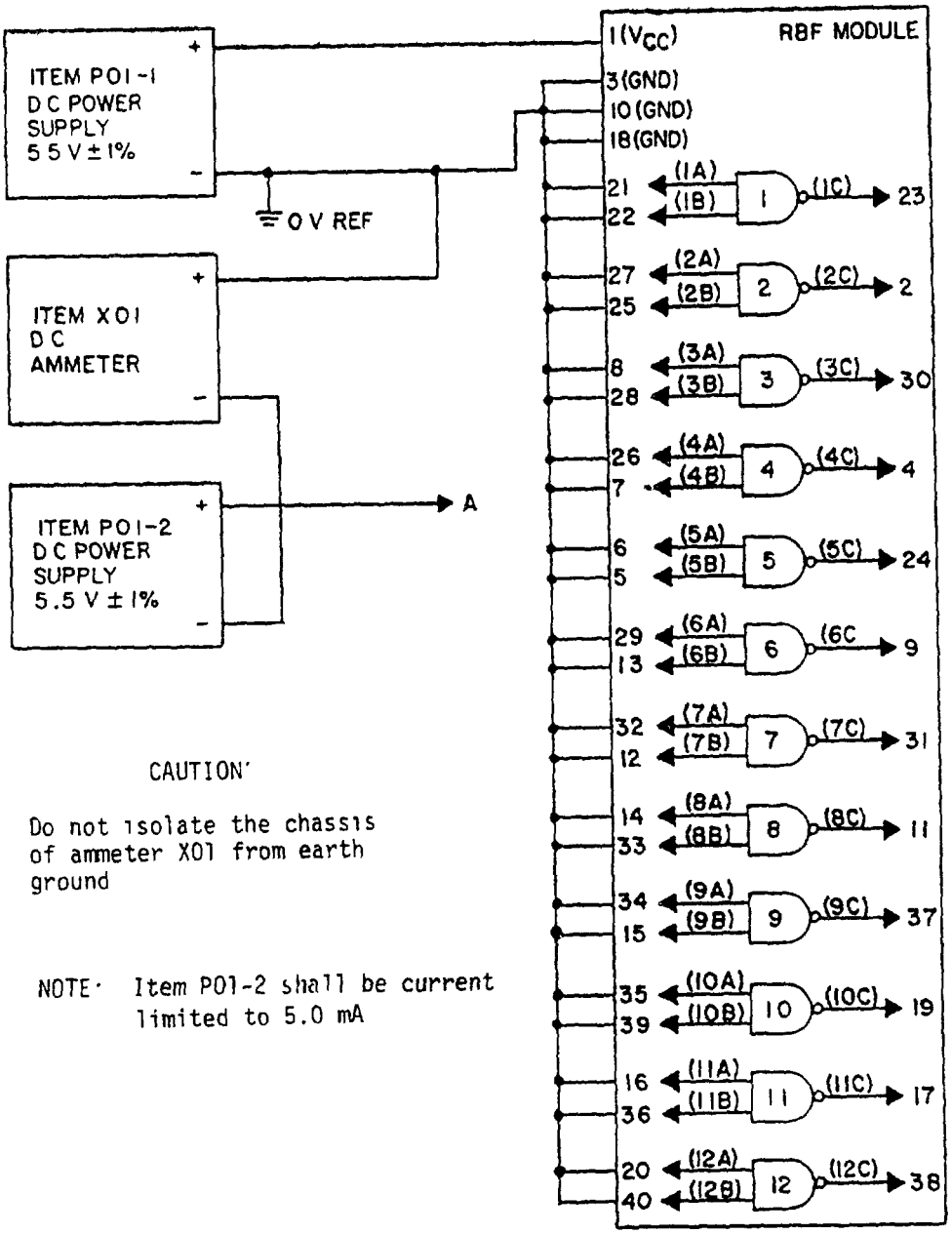


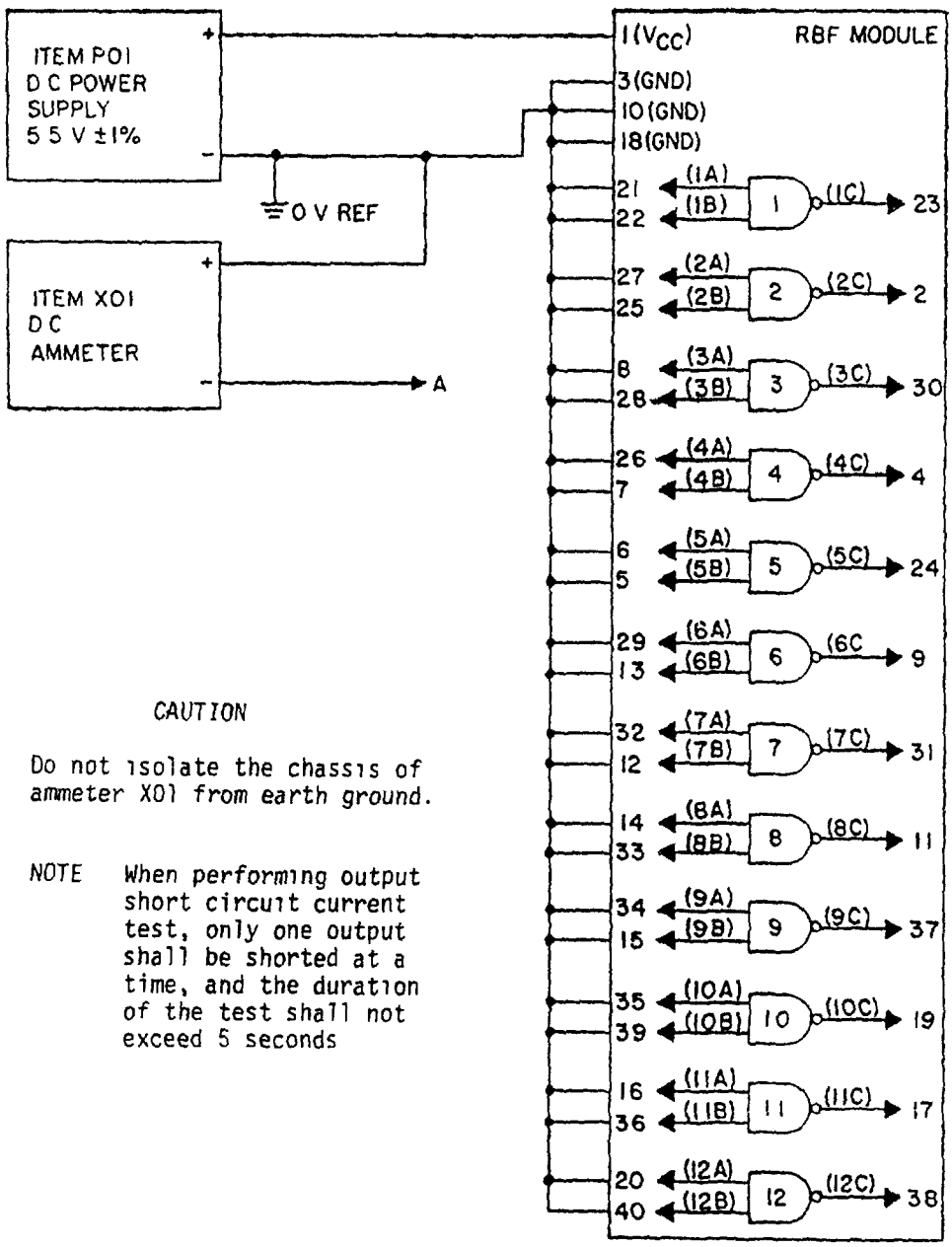
FIGURE 8 Output leakage current test circuit for table VI

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TABLE VII. Output short circuit current test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
E1	1C Output short circuit current	23	-40	-100	mA	9
E2	2C "	2	"	"	"	"
E3	3C "	30	"	"	"	"
E4	4C "	4	"	"	"	"
E5 -	5C "	24	"	"	"	"
E6	6C "	9	"	"	"	"
E7	7C "	31	"	"	"	"
E8	8C "	11	"	"	"	"
E9	9C "	37	"	"	"	"
E10	10C "	19	"	"	"	"
E11	11C "	17	"	"	"	"
E12	12C "	38	"	"	"	"

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CAUTION

Do not isolate the chassis of ammeter X01 from earth ground.

NOTE When performing output short circuit current test, only one output shall be shorted at a time, and the duration of the test shall not exceed 5 seconds

FIGURE 9. Output short circuit current test circuit for table VII.

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TABLE VIII. Input clamp diode voltage test.

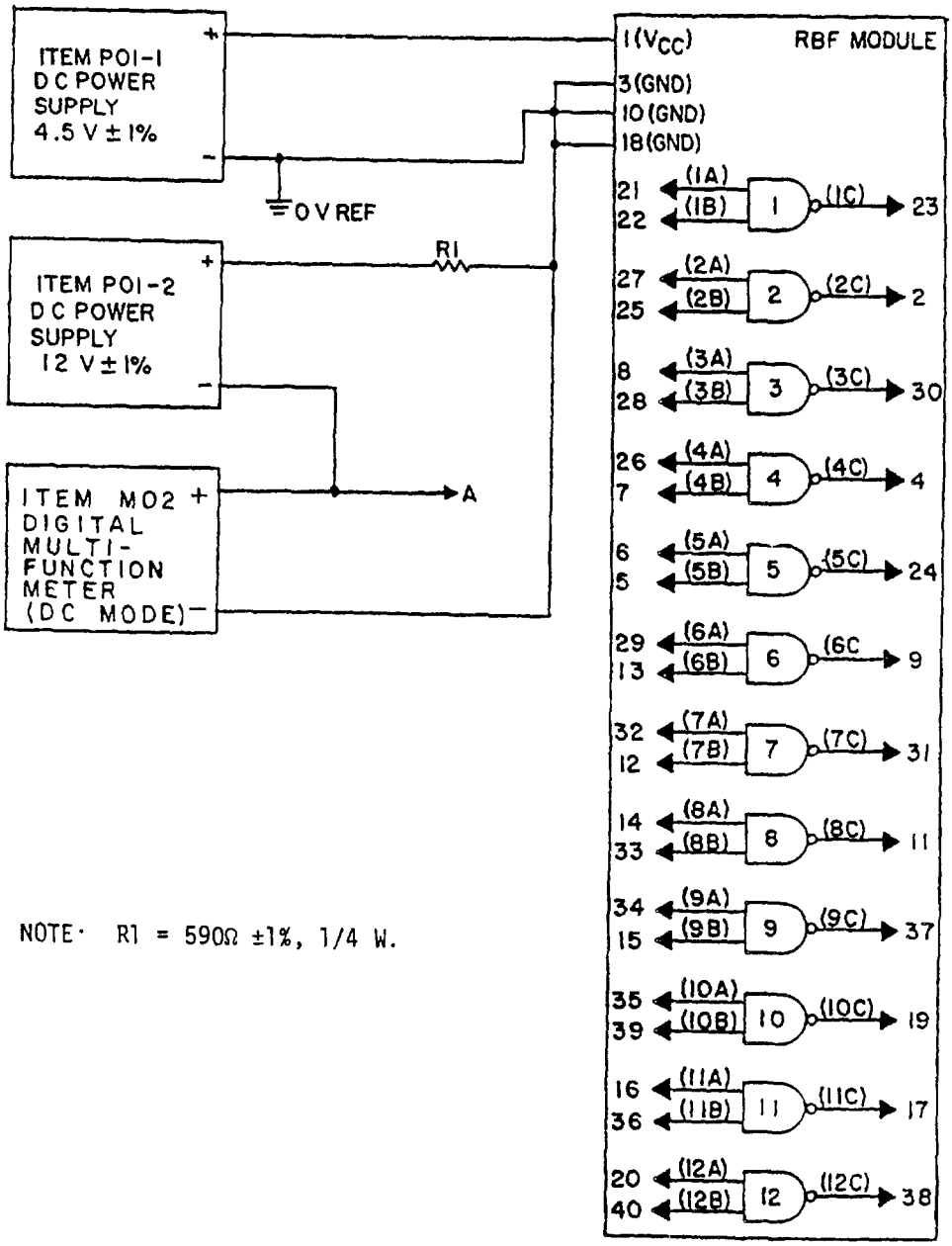
Test condition	Parameter		Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
F1	1A	Input clamp diode voltage	21	-0.4	-1.2	V	10
F2	1B	"	22	"	"	"	"
F3	2A	"	27	"	"	"	"
F4	2B	"	25	"	"	"	"
F5	3A	"	8	"	"	"	"
F6	3B	"	28	"	"	"	"
F7	4A	"	26	"	"	"	"
F8	4B	"	7	"	"	"	"
F9	5A	"	6	"	"	"	"
F10	5B	"	5	"	"	"	"
F11	6A	"	29	"	"	"	"
F12	6B	"	13	"	"	"	"
F13	7A	"	32	"	"	"	"
F14	7B	"	12	"	"	"	"
F15	8A	"	14	"	"	"	"
F16	8B	"	33	"	"	"	"
F17	9A	"	34	"	"	"	"
F18	9B	"	15	"	"	"	"

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TABLE VIII. Input clamp diode voltage test - Continued.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
F19	10A Input clamp diode voltage	35	-0.4	-1.2	V	10
F20	10B "	39	"	"	"	"
F21	11A "	16	"	"	"	"
F22	11B "	36	"	"	"	"
F23	12A "	20	"	"	"	"
F24	12B "	40	"	"	"	"

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NOTE: R1 = 590Ω ±1%, 1/4 W.

* FIGURE 10 Input clamp diode voltage test circuit for table VIII.

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*TABLE IX. Output voltage (low-level) test.

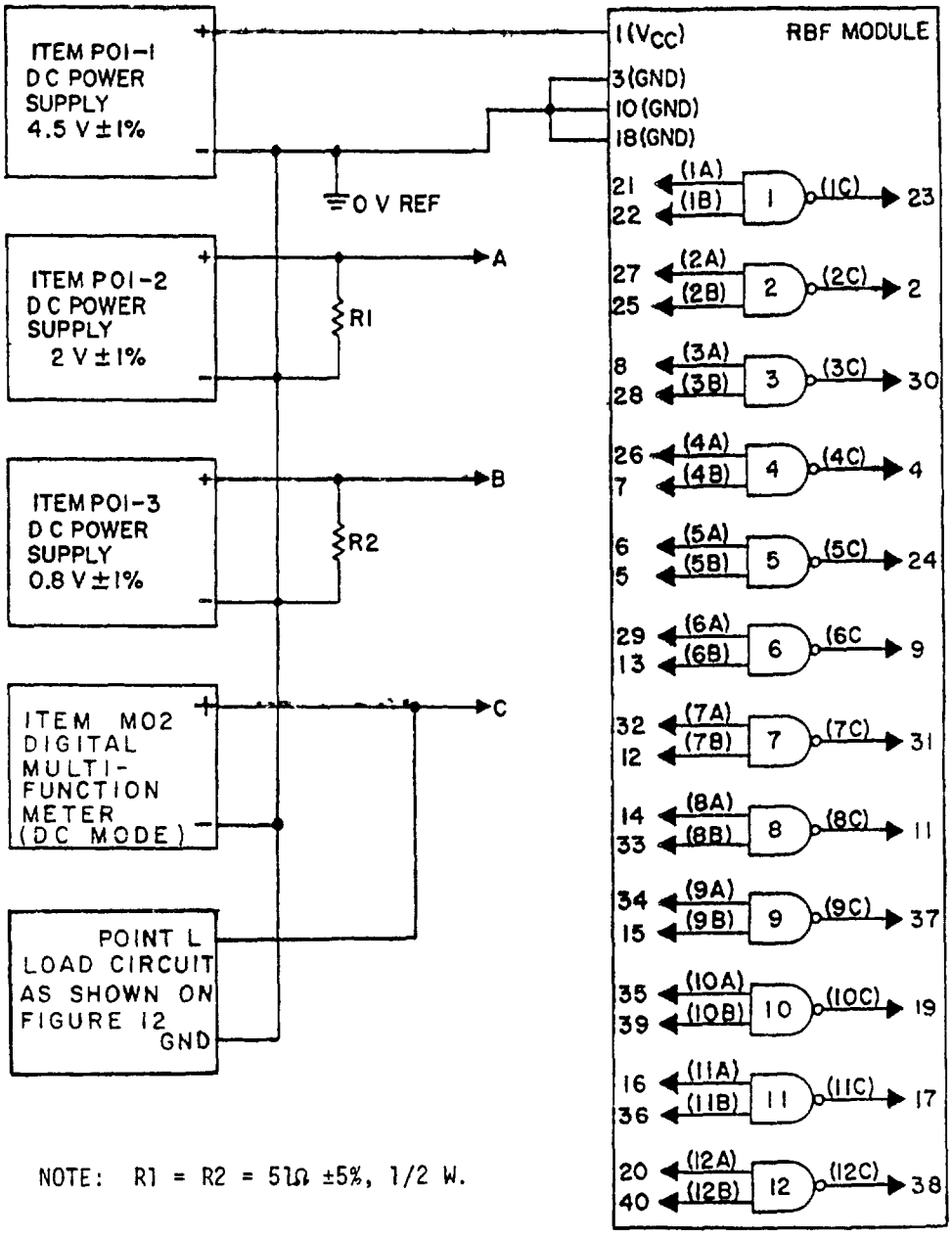
Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
G1	1C	Output voltage, low level	21,22	Open	23	0	500	mV	11
G2	2C	"	25,27	"	2	"	"	"	"
G3	3C	"	8,28	"	30	"	"	"	"
G4	4C	"	7,26	"	4	"	"	"	"
G5	5C	"	5,6	"	24	"	"	"	"
G6	6C	"	13,29	"	9	"	"	"	"
G7	7C	"	12,32	"	31	"	"	"	"
G8	8C	"	14,33	"	11	"	"	"	"
G9	9C	"	15,34	"	37	"	"	"	"
G10	10C	"	35,39	"	19	"	"	"	"
G11	11C	"	16,36	"	17	"	"	"	"
G12	12C	"	20,40	"	38	"	"	"	"

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*TABLE X. Output voltage (high-level) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
H1	1C Output voltage, high level	22	21	23	2.5	4.5	V	11
H2	2C "	25	27	2	"	"	"	"
H3	3C "	28	8	30	"	"	"	"
H4	4C "	7	26	4	"	"	"	"
H5	5C "	5	6	24	"	"	"	"
H6	6C "	13	29	9	"	"	"	"
H7	7C "	12	32	31	"	"	"	"
H8	8C "	33	14	11	"	"	"	"
H9	9C "	15	34	37	"	"	"	"
H10	10C "	39	35	19	"	"	"	"
H11	11C "	36	16	17	"	"	"	"
H12	12C "	40	20	38	"	"	"	"
H13	1C "	21	22	23	"	"	"	"
H14	2C "	27	25	2	"	"	"	"
H15	3C "	8	28	30	"	"	"	"
H16	4C "	26	7	4	"	"	"	"
H17	5C "	6	5	24	"	"	"	"
H18	6C "	29	13	9	"	"	"	"
H19	7C "	32	12	31	"	"	"	"
H20	8C "	14	33	11	"	"	"	"
H21	9C "	34	15	37	"	"	"	"
H22	10C "	35	39	19	"	"	"	"
H23	11C "	16	36	17	"	"	"	"
H24	12C "	20	40	38	"	"	"	"

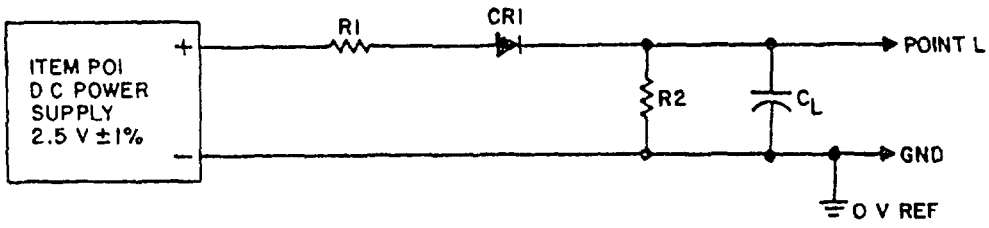
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NOTE: R1 = R2 = 51Ω ±5%, 1/2 W.

* FIGURE 11. Output voltage (low-level and high-level) test circuit for tables IX and X.

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NOTES.
1. CR1 = 1N3600, 1N4150 or equivalent.
2. R1 = Select value for a current flow of 20 mA ±5% out of point L with point L held at 0.5 V.
3. R2 = 2.49 kΩ ±1%, 1/8 W.
4. C_L = 50 ±5 pF (including probe and parasitic capacitance) for test conditions N1-N12, P1-P12, Q1-Q12, R1-R12, S1-S12, T1-T12, U1-U12, V1-V12.
5. C_L = 150 ±5 pF (including probe and parasitic capacitance) for test conditions N13-N24, P13-P24, Q13-Q24, and R13-R24.
6. Capacitor C_L is not required for dc electrical test conditions G1-G12 and H1-H24.
7. Lead length from load circuit to module output pin under test is to be less than 1.00 inch (25.4 mm).

FIGURE 12. RBF logic load circuit.

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TABLE XI. Input current (high-level) test.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
J1	1A	Input current, high level	5,6,7,8,12,13,14,15,16, 20,22,25,26,27,28,29, 32,33,34,35,36,39,40	21	0	50	µA	13
J2	1B	"	5,6,7,8,12,13,14,15,16, 20,21,25,26,27,28,29, 32,33,34,35,36,39,40	22	"	"	"	"
J3	2A	"	5,6,7,8,12,13,14,15,16, 20,21,22,25,26,28,29, 32,33,34,35,36,39,40	27	"	"	"	"
J4	2B	"	5,6,7,8,12,13,14,15,16, 20,21,22,26,27,28,29, 32,33,34,35,36,39,40	25	"	"	"	"
J5	3A	"	5,6,7,12,13,14,15,16, 20,21,22,25,26,27,28, 29,32,33,34,35,36,39,40	8	"	"	"	"
J6	3B	"	5,6,7,8,12,13,14,15,16, 20,21,22,25,26,27,29, 32,33,34,35,36,39,40	28	"	"	"	"
J7	4A	"	5,6,7,8,12,13,14,15,16, 20,21,22,25,27,28,29, 32,33,34,35,36,39,40	26	"	"	"	"
J8	4B	"	5,6,8,12,13,14,15,16, 20,21,22,25,26,27,28, 29,32,33,34,35,36,39,40	7	"	"	"	"
J9	5A	"	5,7,8,12,13,14,15,16, 20,21,22,25,26,27,28, 29,32,33,34,35,36,39,40	6	"	"	"	"
J10	5B	"	6,7,8,12,13,14,15,16, 20,21,22,25,26,27,28, 29,32,33,34,35,36,39,40	5	"	"	"	"
J11	6A	"	5,6,7,8,12,13,14,15,16, 20,21,22,25,26,27,28, 32,33,34,35,36,39,40	29	"	"	"	"

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TABLE XI. Input current (high-level) test - Continued.

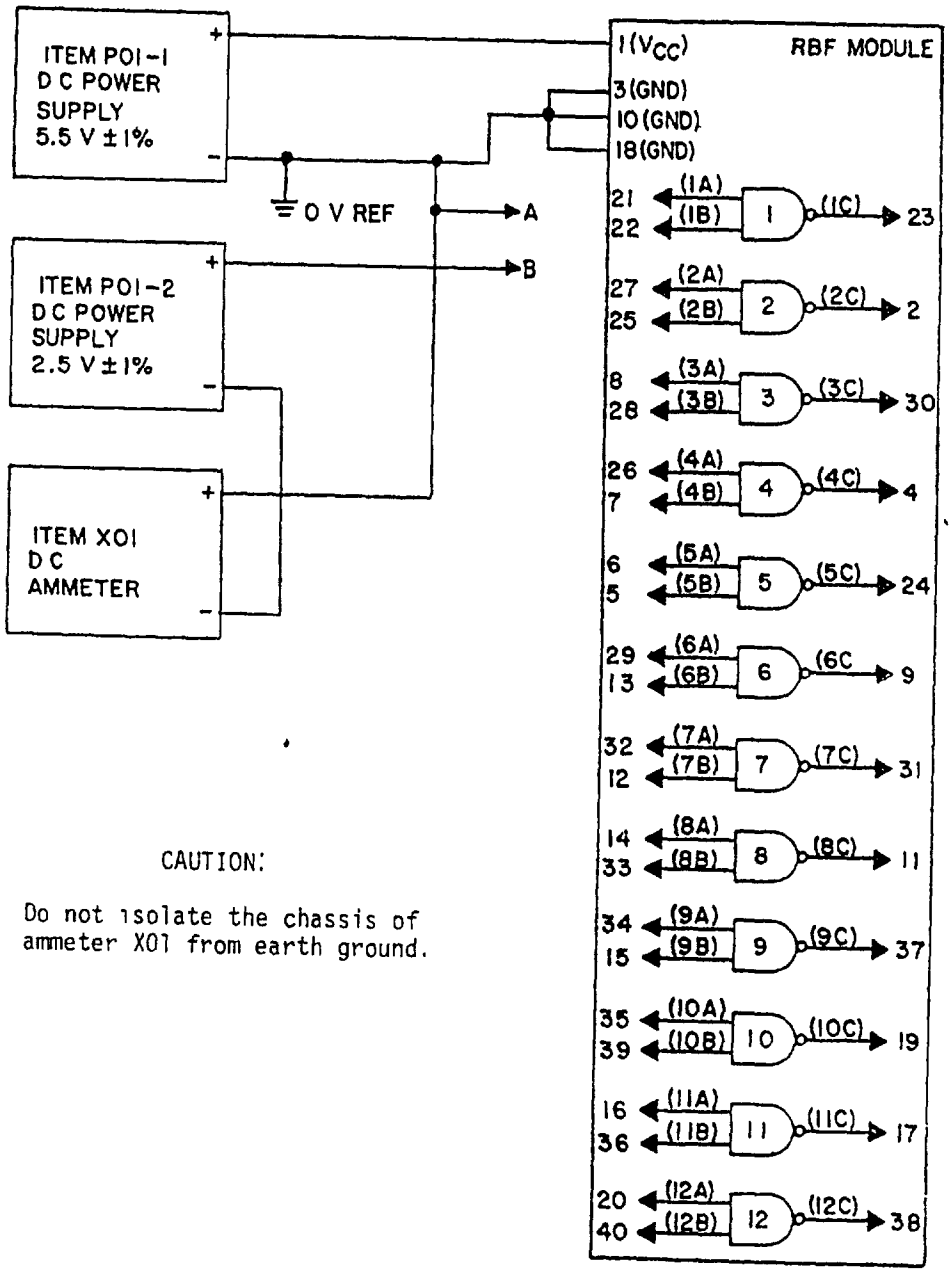
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
J12	6B Input current, high level	5,6,7,8,12,14,15,16,20, 21,22,25,26,27,28,29, 32,33,34,35,36,39,40	13	0	50	µA	13
J13	7A "	5,6,7,8,12,13,14,15,16, 20,21,22,25,26,27,28, 29,33,34,35,36,39,40	32	"	"	"	"
J14	7B "	5,6,7,8,13,14,15,16,20, 21,22,25,26,27,28,29, 32,33,34,35,36,39,40	12	"	"	"	"
J15	8A "	5,6,7,8,12,13,15,16,20, 21,22,25,26,27,28,29, 32,33,34,35,36,39,40	14	0	50	µA	13
J16	8B "	5,6,7,8,12,13,14,15,16, 20,21,22,25,26,27,28, 29,32,34,35,36,39,40	33	"	"	"	"
J17	9A "	5,6,7,8,12,13,14,15,16, 20,21,22,25,26,27,28, 29,32,33,35,36,39,40	34	0	50	µA	13
J18	9B "	5,6,7,8,12,13,14,16, 20,21,22,25,26,27,28, 29,32,33,34,35,36,39,40	15	"	"	"	"
J19	10A "	5,6,7,8,12,13,14,15,16, 20,21,22,25,26,27,28, 29,32,33,34,36,39,40	35	"	"	"	"
J20	10B "	5,6,7,8,12,13,14,15,16, 20,21,22,25,26,27,28, 29,32,33,34,35,36,40	39	"	"	"	"
J21	11A "	5,6,7,8,12,13,14,15,20, 21,22,25,26,27,28,29, 32,33,34,35,36,39,40	16	"	"	"	"
J22	11B "	5,6,7,8,12,13,14,15,16, 20,21,22,25,26,27,28, 29,32,33,34,35,39,40	36	"	"	"	"

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TABLE XI. Input current (high-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
J23	12A Input current, high level	5,6,7,8,12,13,14,15,16, 21,22,25,26,27,28,29, 32,33,34,35,36,39,40	20	0	50	µA	13
J24	12B "	5,6,7,8,12,13,14,15,16, 20,21,22,25,26,27,28, 29,32,33,34,35,36,39	40	"	"	"	"

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CAUTION:

Do not isolate the chassis of ammeter XO1 from earth ground.

FIGURE 13 Input current (high-level) test circuit for table XI

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TABLE XII. Input current (low-level) test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
K1	1A Input current, low level	21	-0.4	-2.2	mA	14
K2	1B "	22	"	"	"	"
K3	2A "	27	"	"	"	"
K4	2B "	25	"	"	"	"
K5	3A "	8	"	"	"	"
K6	3B "	28	"	"	"	"
K7	4A "	26	"	"	"	"
K8	4B "	7	"	"	"	"
K9	5A "	6	"	"	"	"
K10	5B "	5	"	"	"	"
K11	6A "	29	"	"	"	"
K12	6B "	13	"	"	"	"
K13	7A "	32	"	"	"	"
K14	7B "	12	"	"	"	"
K15	8A "	14	"	"	"	"
K16	8B "	33	"	"	"	"
K17	9A "	34	"	"	"	"
K18	9B "	15	"	"	"	"
K19	10A "	35	"	"	"	"
K20	10B "	39	"	"	"	"
K21	11A "	16	"	"	"	"
K22	11B "	36	"	"	"	"

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TABLE XII. Input current (low-level) test - Continued.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
K23	12A Input current, low level	20	-0.4	-2.2	mA	14
K24	12B Input current, low level	40	-0.4	-2.2	mA	14

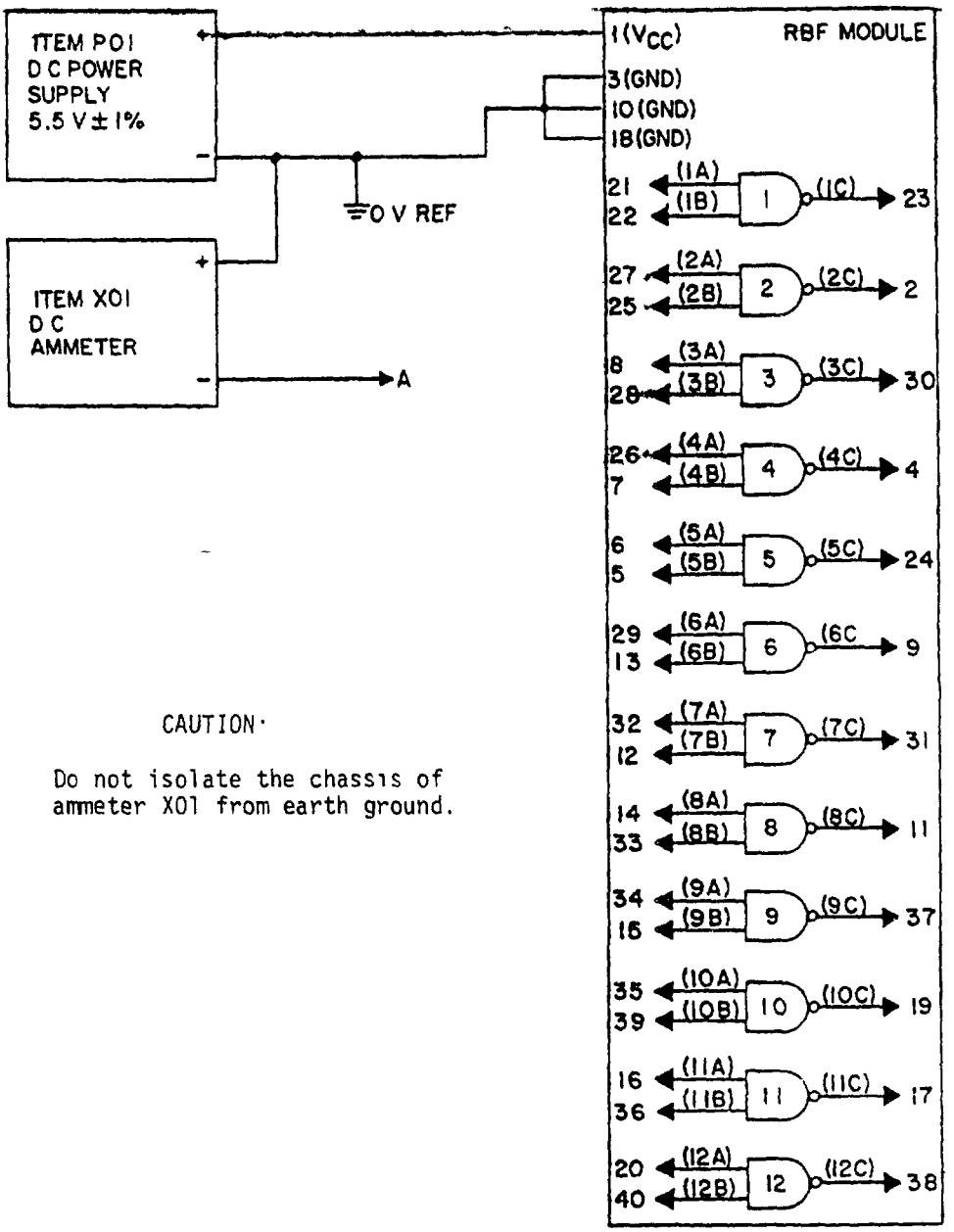
TABLE XIII. Ground pin continuity test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
L1	Ground pin continuity	3	0	1	Ω	15
L2	"	18	0	1	"	"
L3	"	3	0.00	0.10	"	"
L4	"	18	0.00	0.10	"	"

TABLE XIV. Power supply current (low-level and high-level) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to point	B to point	Min	Max		
M1	Power supply current, low level	C	Open	30	108	mA	16
M2	Power supply current, high level	Open	C	15	48	mA	16

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

FIGURE 14. Input current (low-level) test circuit for table XII.

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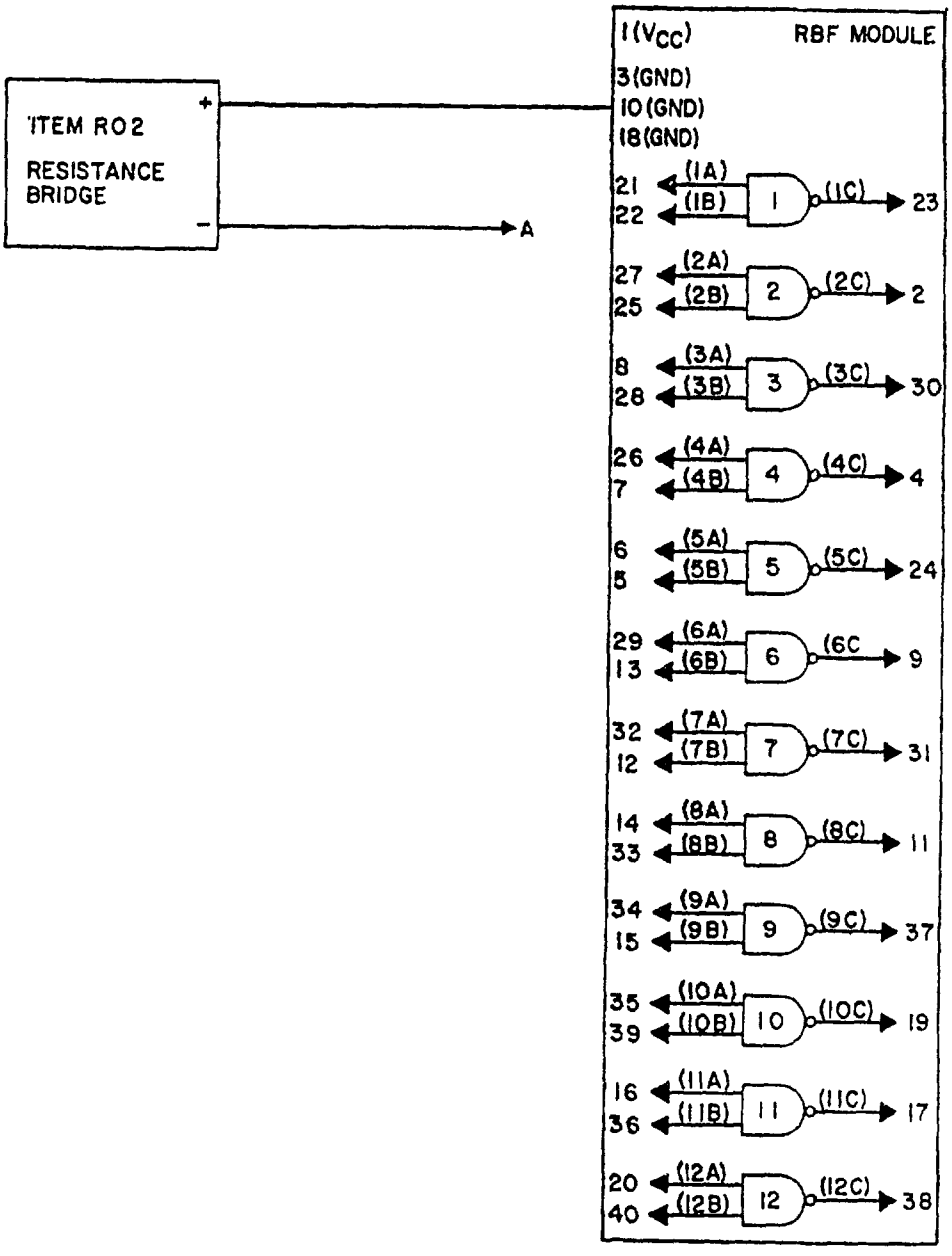


FIGURE 15. Ground pin continuity test circuit for table XIII.

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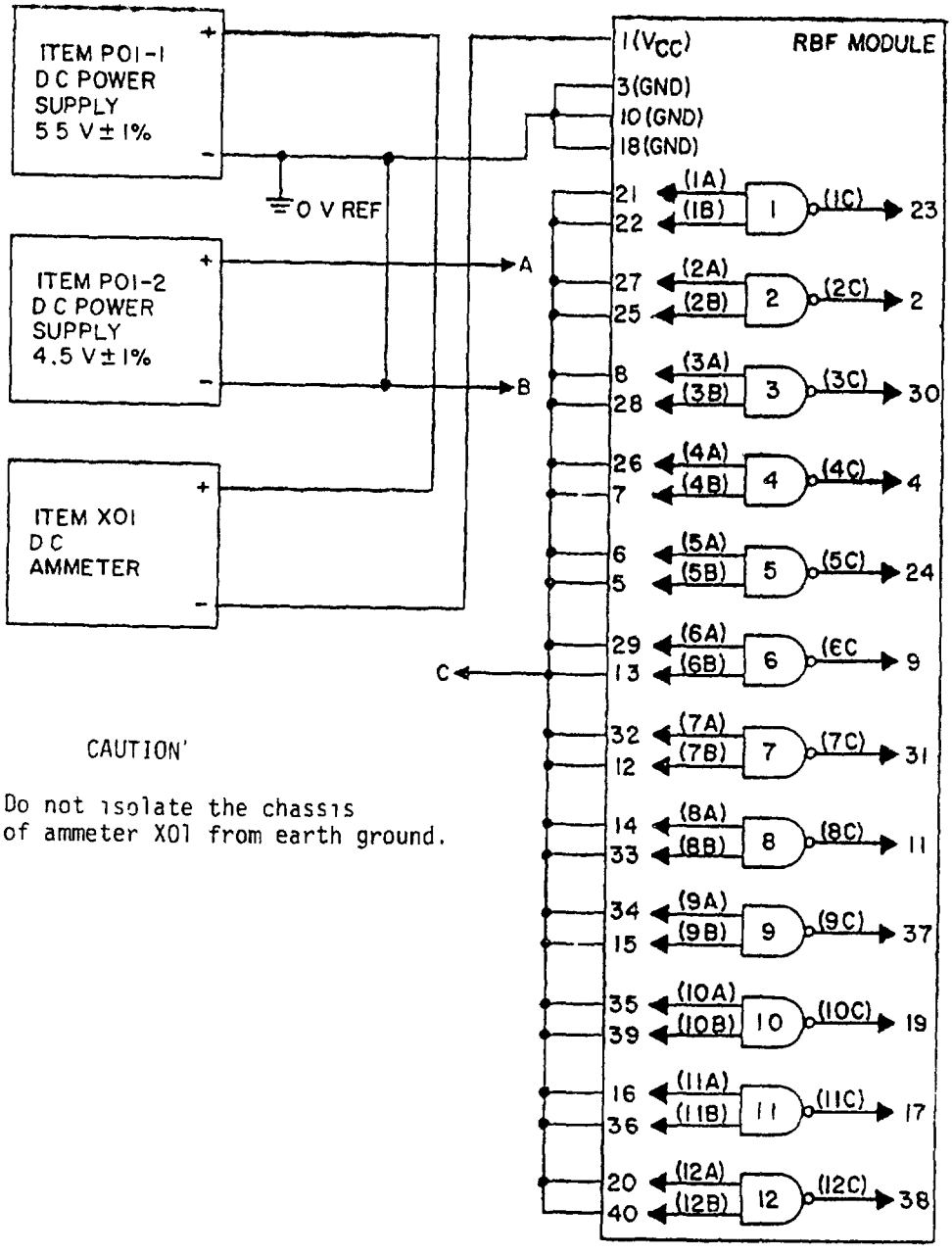


FIGURE 16. Power supply current (low-level and high-level) test circuit for table XIV.

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*TABLE XV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
N1	1C Transition time (t_{TLH}), C_L = 50 pF	21	22	23	Open	2	15	ns	17,18
N2	2C "	27	25	2	"	"	"	"	"
N3	3C "	8	28	30	"	"	"	"	"
N4	4C "	26	7	4	"	"	"	"	"
N5	5C "	6	5	24	"	"	"	"	"
N6	6C "	29	13	9	"	"	"	"	"
N7	7C "	32	12	31	"	"	"	"	"
N8	8C "	14	33	11	"	"	"	"	"
N9	9C "	34	15	37	"	"	"	"	"
N10	10C "	35	39	19	"	"	"	"	"
N11	11C "	16	36	17	"	"	"	"	"
N12	12C "	20	40	38	"	"	"	"	"
N13	1C Transition time (t_{TLH}), C_L = 150 pF	21	22	23	"	"	27	"	"
N14	2C "	27	25	2	"	"	"	"	"
N15	3C "	8	28	30	"	"	"	"	"
N16	4C "	26	7	4	"	"	"	"	"
N17	5C "	6	5	24	"	"	"	"	"
N18	6C "	29	13	9	"	"	"	"	"
N19	7C "	32	12	31	"	"	"	"	"
N20	8C "	14	33	11	"	"	"	"	"

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TABLE IV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
N21	9C Transition time (t_{TLH}), C_L = 150 pF	34	15	37	Open	2	27	ns	17,18
N22	10C "	35	39	19	"	"	"	"	"
N23	11C "	16	36	17	"	"	"	"	"
N24	12C "	20	40	38	"	"	"	"	"
P1	1C Transition time (t_{THL}), C_L = 50 pF	22	21	23	5,6,7,8,12,13, 14,15,16,20, 25,26,27,28, 29,32,33,34, 35,36,39,40	"	6	"	"
P2	2C "	25	27	2	5,6,7,8,12,13, 14,15,16,20, 21,22,26,28, 29,32,33,34, 35,36,39,40	"	"	"	"
P3	3C "	28	8	30	5,6,7,12,13, 14,15,16,20, 21,22,25,26, 27,29,32,33, 34,35,36,39,40	"	"	"	"
P4	4C "	7	26	4	5,6,8,12,13, 14,15,16,20, 21,22,25,27, 28,29,32,33, 34,35,36,39,40	"	"	"	"
P5	5C "	5	6	24	7,8,12,13,14, 15,16,20,21, 22,25,26,27, 28,29,32,33, 34,35,36,39,40	"	"	"	"
P6	6C "	13	29	9	5,6,7,8,12,14 15,16,20,21, 22,25,26,27, 28,32,33,34, 35,36,39,40	"	"	"	"

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*TABLE XV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P7	7C Transition time (t_{THL}), C_L = 50 pF	12	32	31	5,6,7,8,13,14, 15,16,20,21, 22,25,26,27, 28,29,33,34, 35,36,39,40	2	6	ns	17,18
P8	8C "	33	14	11	5,6,7,8,12,13, 15,16,20,21, 22,25,26,27, 28,29,32,34, 35,36,39,40	"	"	"	"
P9	9C "	15	34	37	5,6,7,8,12,13, 14,16,20,21, 22,25,26,27, 28,29,32,33 35,36,39,40	"	"	"	"
P10	10C "	39	35	19	5,6,7,8,12,13, 14,15,16,20, 21,22,25,26, 27,28,29,32 33,34,36,40	"	"	"	"
P11	11C "	36	16	17	5,6,7,8,12,13, 14,15,20,21, 22,25,26,27, 28,29,32,33 34,35,39,40	"	"	"	"
P12	12C "	40	20	38	5,6,7,8,12,13, 14,15,16,21, 22,25,26,27, 28,29,32,33 34,35,36,39	"	"	"	"
P13	1C Transition time (t_{THL}), C_L = 150 pF	22	21	23	5,6,7,8,12,13, 14,15,16,20, 25,26,27,28, 29,32,33,34 35,36,39,40	"	11	"	"

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*TABLE XV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P14	2C Transition time (t_{THL}), C_L = 150 pF	25	27	2	5,6,7,8,12,13, 14,15,16,20, 21,22,26,28, 29,32,33,34, 35,36,39,40	2	11	ns	17,18
P15	3C "	28	8	30	5,6,7,12,13, 14,15,16,20, 21,22,25,26, 27,29,32,33, 34,35,36,39,40	"	"	"	"
P16	4C "	7	26	4	5,6,8,12,13, 14,15,16,20, 21,22,25,27, 28,29,32,33, 34,35,36,39,40	"	"	"	"
P17	5C "	5	6	24	7,8,12,13,14, 15,16,20,21, 22,25,26,27, 28,29,32,33, 34,35,36,39,40	"	"	"	"
P18	6C "	13	29	9	5,6,7,8,12,14, 15,16,20,21, 22,25,26,27, 28,32,33,34, 35,36,39,40	"	"	"	"
P19	7C "	12	32	31	5,6,7,8,13,14, 15,16,20,21, 22,25,26,27, 28,29,33,34, 35,36,39,40	"	"	"	"
P20	8C "	33	14	11	5,6,7,8,12,13, 15,16,20,21, 22,25,26,27, 28,29,32,34, 35,36,39,40	"	"	"	"

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*TABLE XV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition		Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
P21	9C	Transition time (t_{THL}), C_L = 150 pF	15	34	37	5,6,7,8,12,13, 14,16,20,21, 22,25,26,27, 28,29,32,33, 35,36,39,40	2	11	ns	17,18
P22	10C	"	39	35	19	5,6,7,8,12,13, 14,15,16,20,21 21,22,25,26, 27,28,29,32, 33,34,36,40	"	"	"	"
P23	11C	"	36	16	17	5,6,7,8,12,13, 14,15,20,21, 22,25,26,27, 28,29,32,33, 34,35,36,39,40	"	"	"	"
P24	12C	"	40	20	38	5,6,7,8,12,13, 14,15,16,21, 22,25,26,27, 28,29,32,33, 34,35,36,39	"	"	"	"
Q1	1C	Propagation delay time (t_{PLH}), C_L = 50 pF	21	22	23	Open	"	7	"	"
Q2	2C	"	27	25	2	"	"	"	"	"
Q3	3C	"	8	28	30	"	"	"	"	"
Q4	4C	"	26	7	4	"	"	"	"	"
Q5	5C	"	6	5	24	"	"	"	"	"
Q6	6C	"	29	13	9	"	"	"	"	"
Q7	7C	"	32	12	31	"	"	"	"	"
Q8	8C	"	14	33	11	"	"	"	"	"

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*TABLE XV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pir	D to pin	Min	Max		
Q9	9C Propagation delay time (t_{PLH}), C_L = 50 pF	34	15	37	Open	2	7	ns	17,18
Q10	10C "	35	39	19	"	"	"	"	"
Q11	11C "	16	36	17	"	"	"	"	"
Q12	12C "	20	40	38	"	"	"	"	"
Q13	1C Propagation delay time (t_{PLH}), C_L = 150 pF	21	22	23	"	4.0	10.5	"	"
Q14	2C "	27	25	2	"	"	"	"	"
Q15	3C "	8	28	30	"	"	"	"	"
Q16	4C "	26	7	4	"	"	"	"	"
Q17	5C "	6	5	24	"	"	"	"	"
Q18	6C "	29	13	9	"	"	"	"	"
Q19	7C "	32	12	31	"	"	"	"	"
Q20	8C "	14	33	11	"	"	"	"	"
Q21	9C "	34	15	37	"	"	"	"	"
Q22	10C "	35	39	19	"	"	"	"	"
Q23	11C "	16	36	17	"	"	"	"	"
Q24	12C "	20	40	38	"	"	"	"	"
R1	1C Propagation delay time (t_{PHL}), C_L = 50 pF	22	21	23	"	2.0	7.5	"	"
R2	2C Propagation delay time (t_{PHL}), C_L = 50 pF	25	27	2	"	2.0	7.5	"	"

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*TABLE XV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

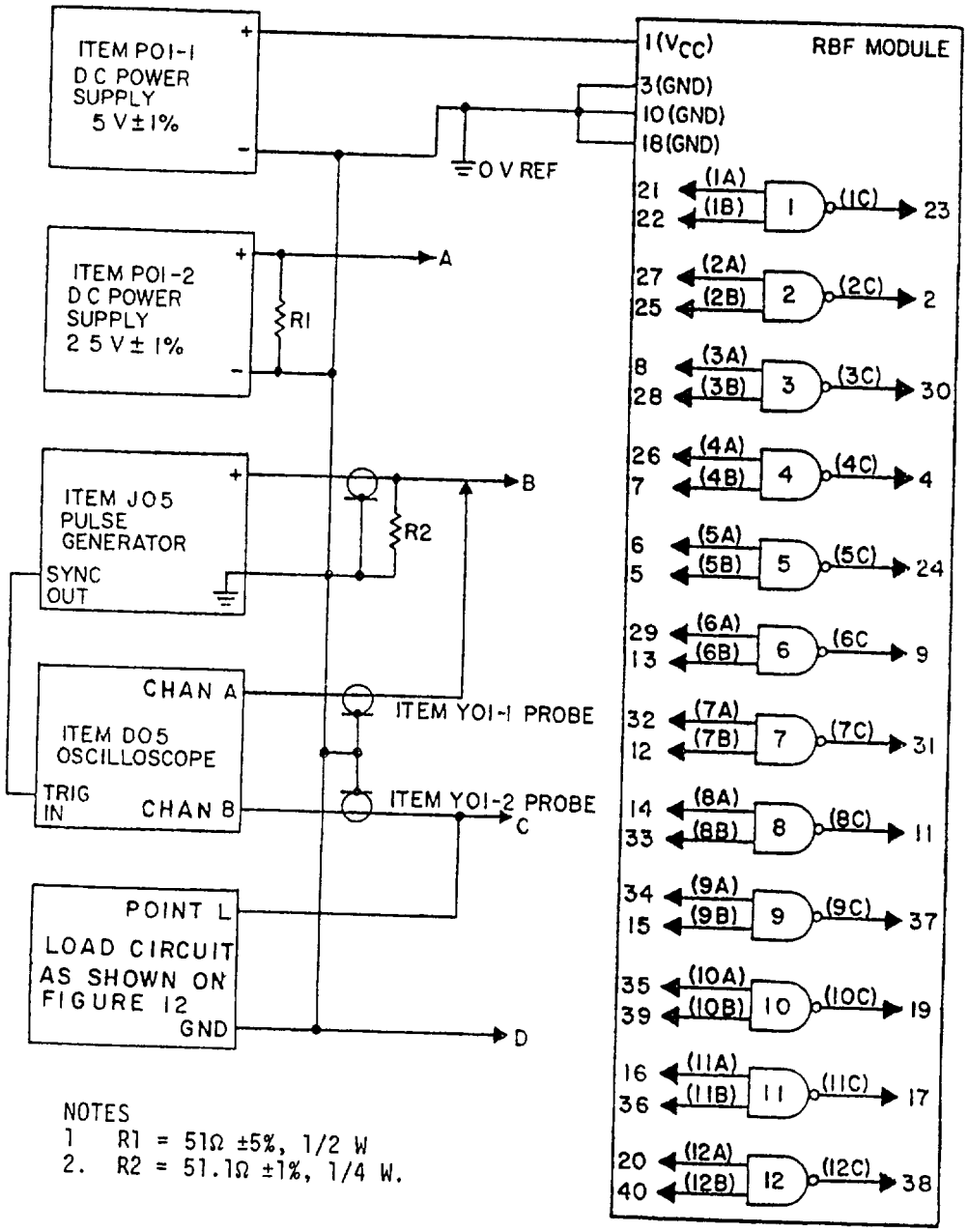
Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R3	3C Propagation delay time (t_{PHL}), C_L = 50 pF	28	8	30	Open	2.0	7.5	ns	17,18
R4	4C "	7	26	4	"	"	"	"	"
R5	5C "	5	6	24	"	"	"	"	"
R6	6C "	13	29	9	"	"	"	"	"
R7	7C "	12	32	31	"	"	"	"	"
R8	8C "	33	14	11	"	"	"	"	"
R9	9C "	15	34	37	"	"	"	"	"
R10	10C "	39	35	19	"	"	"	"	"
R11	11C "	36	16	17	"	"	"	"	"
R12	12C "	40	20	38	"	"	"	"	"
R13	1C Propagation delay time (t_{PHL}), C_L = 150 pF	22	21	23	"	4.0	11.5	"	"
R14	2C "	25	27	2	"	"	"	"	"
R15	3C "	28	8	30	"	"	"	"	"
R16	4C "	7	26	4	Open	"	"	"	"
R17	5C "	5	6	24	"	"	"	"	"
R18	6C "	13	29	9	"	"	"	"	"
R19	7C "	12	32	31	"	"	"	"	"
R20	8C "	33	14	11	"	"	"	"	"
R21	9C "	15	34	37	"	"	"	"	"
R22	10C "	39	35	19	"	"	"	"	"

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*TABLE XV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R23	11C Propagation delay time (t_{PHL}), C_L = 150 pF	36	16	17	Open	4.0	11.5	ns	17,18
R24	12C Propagation delay time (t_{PHL}), C_L = 150 pF	40	20	28	Open	4.0	11.5	ns	17,18

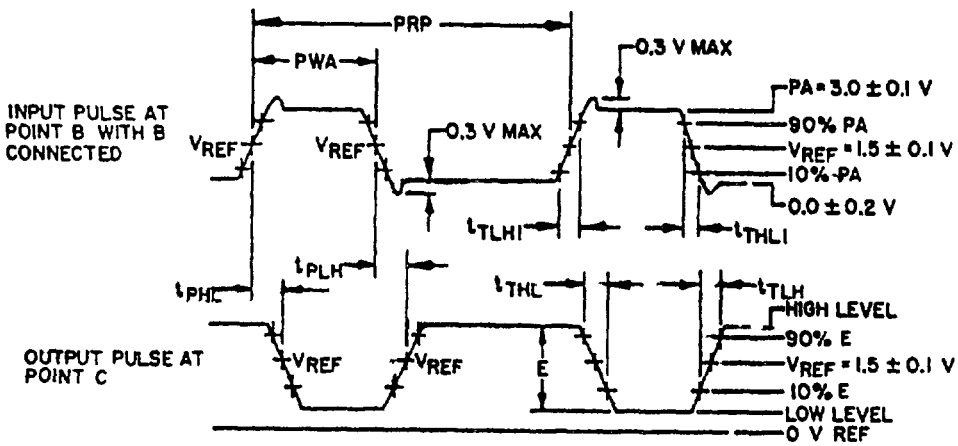
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NOTES
1. R1 = 51Ω ±5%, 1/2 W
2. R2 = 51.1Ω ±1%, 1/4 W.

* FIGURE 17. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for table XV.

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NOTES

1. Input pulse parameters:
   $t_{TLH1} = t_{THL1} = 4 \pm 1$ ns.
   PWA = 100 ±10 ns.
   PRP = 500 ±50 ns.
2. Before making the measurements verify that the output pulse high level is $\geq$ 2.5 V and the low level is $\leq$ 0.5 V.
3. The lead length of electrical conductors between resistor R2 and input pins; probes YO1-1, YO1-2 to input and output pins and between load circuit and output pin shall be less than 1.00 inch (25.4 mm).
4. All voltages are referenced to ground.

* FIGURE 18. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) circuit waveforms for table XV.

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*TABLE XVI. Output to output delta propagation delay time (Δt_{PLH} and Δt_{PHL}) test.

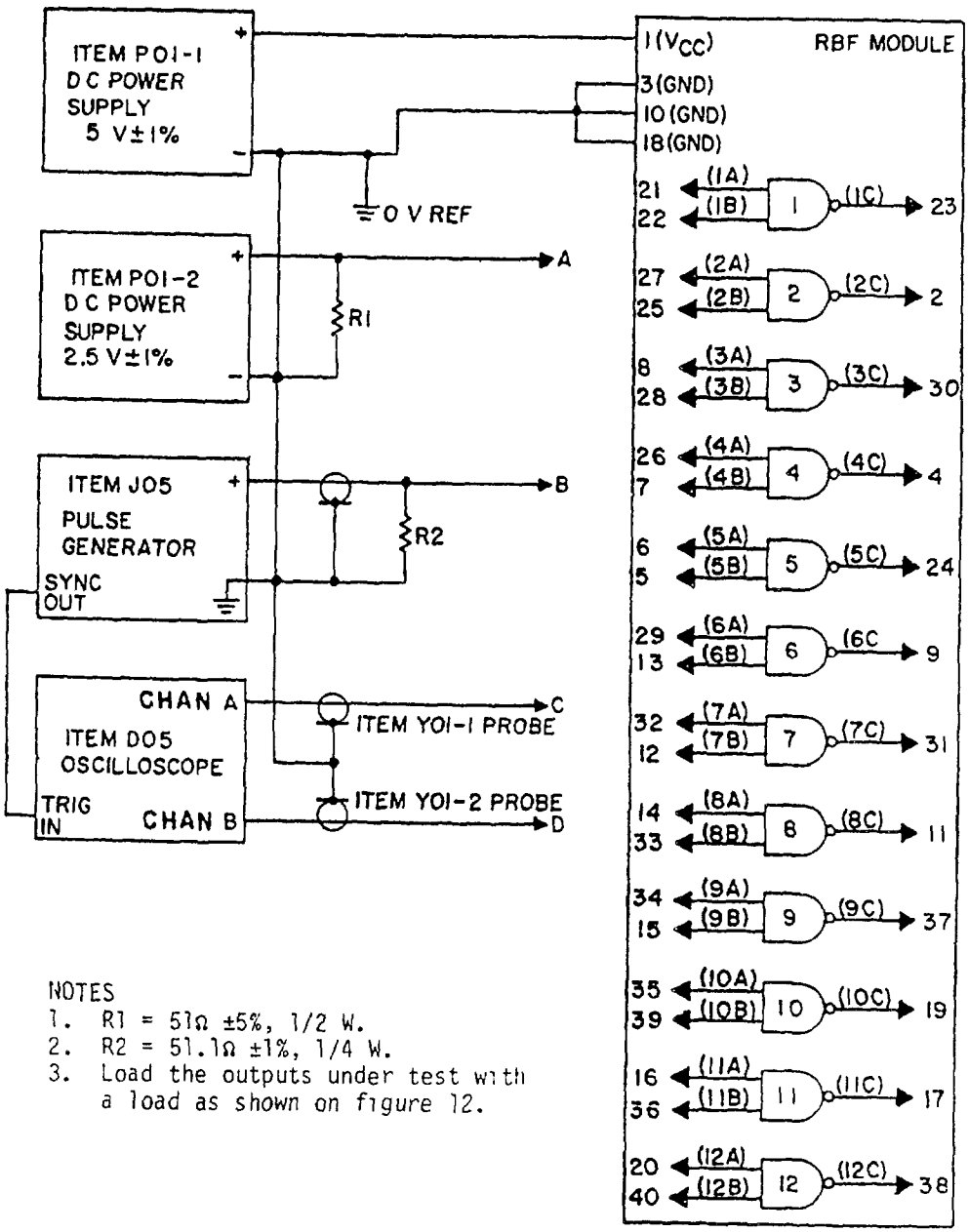
Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S1	1C to 2C Output to output delta propagation delay time (Δt_{PLH}), C_L = 50 pF	8,21,27	22,25,28	23	2	-3	3	ns	19,20
S2	2C to 3C "	8,21,27	22,25,28	2	30	"	"	"	"
S3	1C to 3C "	8,21,27	22,25,28	23	30	"	"	"	"
S4	4C to 5C "	6,26,29	5,7,13	4	24	"	"	"	"
S5	5C to 6C "	6,26,29	5,7,13	24	9	"	"	"	"
S6	4C to 6C "	6,26,29	5,7,13	4	9	"	"	"	"
S7	7C to 8C "	14,32,34	12,15,33	31	11	"	"	"	"
S8	8C to 9C "	14,32,34	12,15,33	11	37	"	"	"	"
S9	7C to 9C "	14,32,34	12,15,33	31	37	"	"	"	"
S10	10C to 11C "	16,20,35	36,39,40	19	17	"	"	"	"
S11	11C to 12C "	16,20,35	36,39,40	17	38	"	"	"	"
S12	10C to 12C "	16,20,35	36,39,40	19	38	"	"	"	"
T1	1C to 12C Output to output delta propagation delay time (Δt_{PHL}), C_L = 50 pF	8,21,27	22,25,28	23	2	"	"	"	"
T2	2C to 3C "	8,21,27	22,25,28	2	30	"	"	"	"
T3	1C to 3C "	8,21,27	22,25,28	23	30	"	"	"	"
T4	4C to 5C "	6,26,29	5,7,13	4	24	"	"	"	"
T5	5C to 6C "	6,26,29	5,7,13	24	9	"	"	"	"
T6	4C to 6C "	6,26,29	5,7,13	4	9	"	"	"	"
T7	7C to 8C "	14,32,34	12,15,33	31	11	"	"	"	"

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*TABLE XVI. Output to output delta propagation delay time (Δt_{PLH} and Δt_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
T8	8C to 9C Output to output delta propagation delay time (Δt_{PHL}), C_L = 50 pF	14,32,34	12,15,33	11	37	-3	3	ns	19,20
T9	7C to 9C "	14,32,34	12,15,33	31	37	"	"	"	"
T10	10C to 11C "	16,20,35	36,39,40	19	17	"	"	"	"
T11	11C to 12C "	16,20,35	36,39,40	17	38	"	"	"	"
T12	10C to 12C "	16,20,35	36,39,40	19	38	"	"	"	"

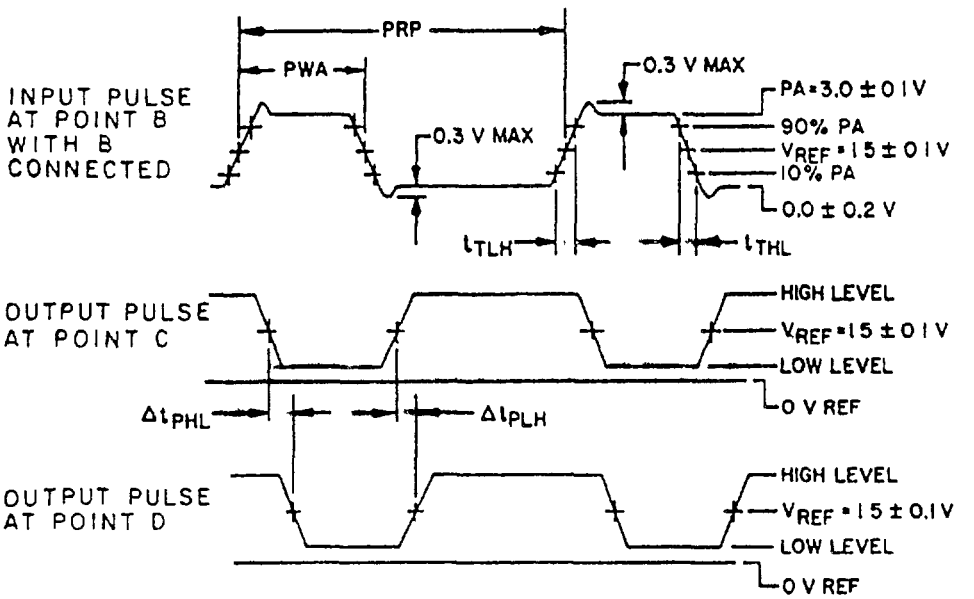
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NOTES
1. R1 = 51Ω ±5%, 1/2 W.
2. R2 = 51.1Ω ±1%, 1/4 W.
3. Load the outputs under test with a load as shown on figure 12.

* FIGURE 19. Output to output delta propagation delay time (Δt_{PLH} and Δt_{PHL}) test circuit for table XVI.

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NOTES
1. Input pulse parameters:
   $t_{TLH} = t_{THL} = 4 \pm 1$ ns.
   PWA = 100 ±10 ns
   PRP = 500 ±50 ns.
2. Before making the measurements specified in table XVI, verify that the output pulse high level is $\geq$ 2.5 V and the low level is $\leq$ 0.5 V.
3. The lead length of electrical conductors between resistor R2 and input pins, probes YO1-1, YO1-2 to output pins, and between load circuits and output pins shall be less than 1.00 inch (25.4 mm).
4. All voltages are referenced to ground.

* FIGURE 20. Output to output delta propagation delay time (Δt_{PLH} and Δt_{PHL}) test circuit waveforms for table XVI.

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TABLE XVII. Output crosstalk (low-level and high-level) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
U1	1C Output crosstalk, low level	5,7,12, 13,15,21, 22,25,28, 33,36,39, 40	Open	6,8,14, 16,20,26 27,29,32, 34,35	23	0	500	mV	21,22
U2	2C "	5,7,12, 13,15,22, 25,27,28, 33,36,39, 40	"	6,8,14, 16,20,21, 26,29,32, 34,35	2	"	"	"	"
U3	3C "	5,7,8,12, 13,15,22, 25,28,33, 36,39,40	"	6,14,16, 20,21,26, 27,29,32, 34,35	30	"	"	"	"
U4	4C "	5,7,12,13, 15,22,25, 26,28,33, 36,39,40	"	6,8,14, 16,20,21, 27,29,32, 34,35	4	"	"	"	"
U5	5C "	5,6,7,12, 13,15,22, 25,28,33, 36,39,40	"	8,14,16, 20,21,26, 27,29,32, 34,35	24	"	"	"	"
U6	6C "	5,7,12,13, 15,22,25, 28,29,33, 36,39,40	"	6,8,14, 16,20,21, 26,27,32, 34,35	9	"	"	"	"
U7	7C "	5,7,12,13, 15,22,25, 28,32,33, 36,39,40	"	6,8,14, 16,20,21, 26,27,29, 34,35	31	"	"	"	"
U8	8C "	5,7,12,13, 14,15,22, 25,28,33, 36,39,40	"	6,8,16, 20,21,26, 27,29,32, 34,35	11	"	"	"	"

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TABLE XVII. Output crosstalk (low-level and high-level) test - Continued.

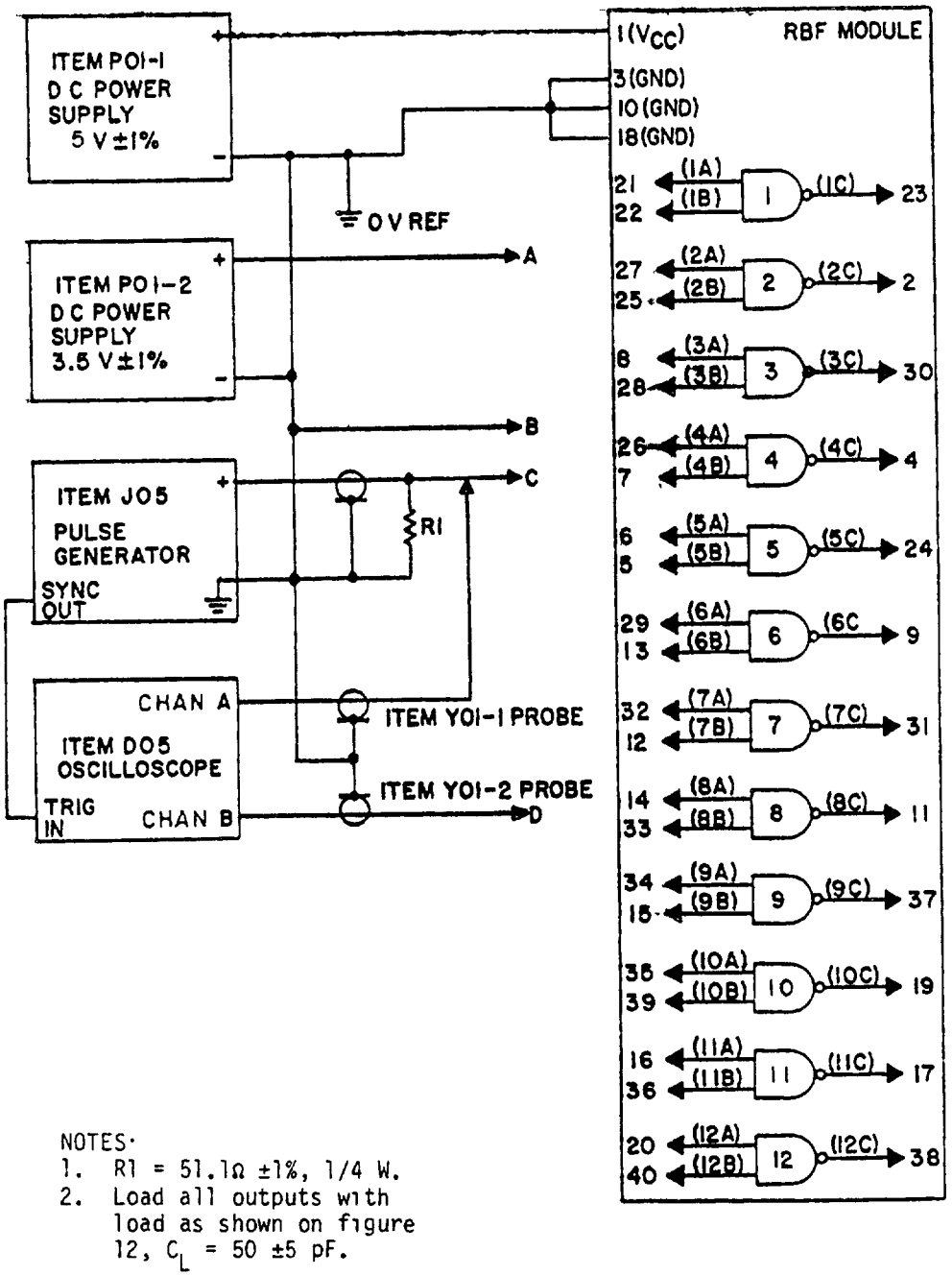
Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
U9	9C Output crosstalk, low level	5,7,12,13, 15,22,25, 28,33,34, 36,39,40	Open	6,8,14,16, 20,21,26, 27,29,32,35	37	0	500	mV	21,22
U10	10C "	5,7,12,13, 15,22,25, 28,33,35, 36,39,40		6,8,14,16, 20,21,26,27, 29,32,34	19	"	"	"	"
U11	11C "	5,7,12, 13,15,16, 22,25,28, 33,36,39, 40	Open	6,8,14, 20,21,26, 27,29,32, 34,35	17	"	"	"	"
U12	12C "	5,7,12,13, 15,20,22, 25,28,33, 36,39,40	Open	6,8,14, 16,21,26, 27,29,32, 34,35	38	"	"	"	"
V1	1C Output crosstalk, high level	5,7,12,13, 15,25,28, 33,36,39, 40	21,22	6,8,14, 16,20,26, 27,29,32, 34,35	23	"	400	"	"
V2	2C "	5,7,12,13, 15,22,28, 33,36,39, 40	25,27	6,8,14, 16,20,21, 26,29,32, 34,35	2	"	"	"	"
V3	3C "	5,7,12,13, 15,22,25, 33,36,39, 40	8,28	6,14,16, 20,21,26, 27,29,32, 34,35	30	"	"	"	"
V4	4C "	5,12,13, 15,22,25, 28,33,36, 39,40	7,26	6,8,14, 16,20,21, 27,29,32, 34,35	4	"	"	"	"

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TABLE XVII. Output crosstalk (low-level and high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
V5	5C Output crosstalk, high level	7,12,13, 15,22,25, 28,33,36, 39,40	5,6	8,14,16, 20,21,26, 27,29,32, 34,35	24	0	400	mV	21,22
V6	6C "	5,7,12, 15,22,25, 28,33,36, 39,40	13,29	6,8,14,16, 20,21,26, 27,32,34, 35	9	"	"	"	"
V7	7C "	5,7,13, 15,22,25, 28,33,36, 39,40	12,32	6,8,14, 16,20,21, 26,27,29, 34,35	31	"	"	"	"
V8	8C "	5,7,12,13, 15,22,25, 28,36,39, 40	14,33	6,8,16, 20,21,26, 27,29,32, 34,35	11	"	"	"	"
V9	9C "	5,7,12,13, 22,25,28, 33,36,39, 40	15,34	6,8,14, 16,20,21, 26,27,29, 32,35	37	"	"	"	"
V10	10C "	5,7,12,13, 15,22,25, 28,33,36, 40	35,39	6,8,14, 16,20,21, 26,27,29, 32,34	19	"	"	"	"
V11	11C "	5,7,12,13, 15,22,25, 28,33,39, 40	16,36	6,8,14, 20,21,26, 27,29,32, 34,35	17	"	"	"	"
V12	12C "	5,7,12,13, 15,22,25, 28,33,36, 39	20,40	6,8,14, 16,21,26, 27,29,32, 34,35	38	"	"	"	"

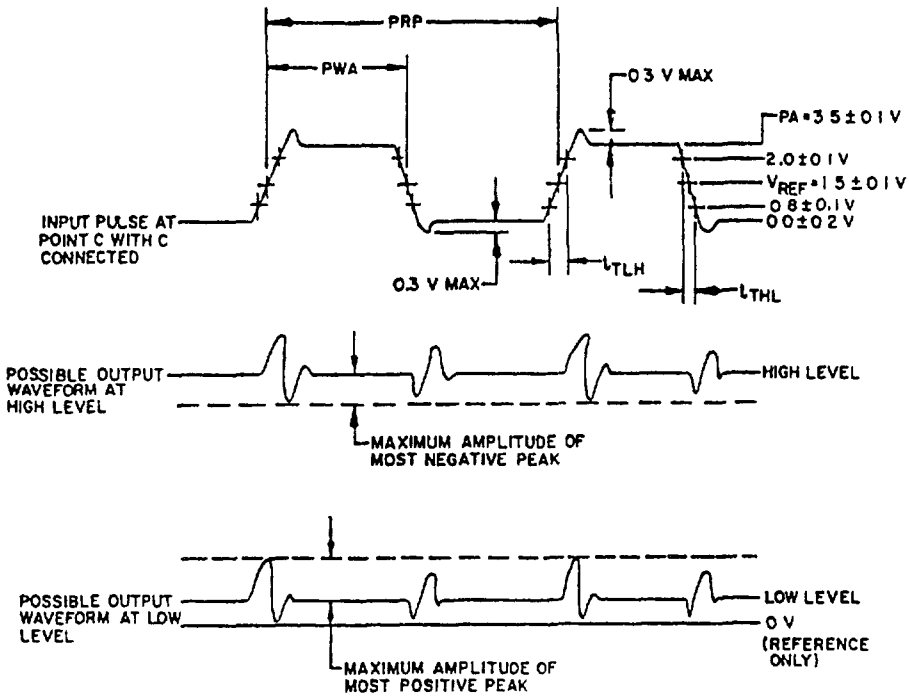
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NOTES:
1. R1 = 51.1Ω ±1%, 1/4 W.
2. Load all outputs with load as shown on figure 12, C_L = 50 ±5 pF.

FIGURE 21. Output crosstalk (low-level and high-level) test circuit for table XVII.

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NOTES

1 Input pulse parameters: For each test condition with the module connected to the pulse generator, adjust the pulse generator for the required pulse amplitude, pulse low level voltage, and transition times. Verify the input pulse parameters at each input pin before making each measurement.

$t_{THL} = t_{TLH} = 3.5 \pm 1.5$ ns (measured at the 0.8 V and 2.0 V points of the input pulse).
PWA = 500 ±50 ns
PRP = 1,000 ±100 ns.

2 The lead length of electrical conductors between resistor R1 and input pins, probes Y01-1, Y01-2 to input and output pins; load circuits to output pins; and ground leads of probes Y01-1, Y01-2 to test fixture and module ground pins shall be less than 1.00 inch (25.4 mm).

FIGURE 22. Output crosstalk (low-level and high-level) test circuit waveforms for table XVII.

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TABLE XVIII. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
W	Power supply transient amplitude	0	600	mV	23,24

TABLE XIX. Power supply filter capacitance test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
X	Power supply filter capacitance	Open	1	1.6	3.0	µF	25

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TABLE XX. Input capacitance test.

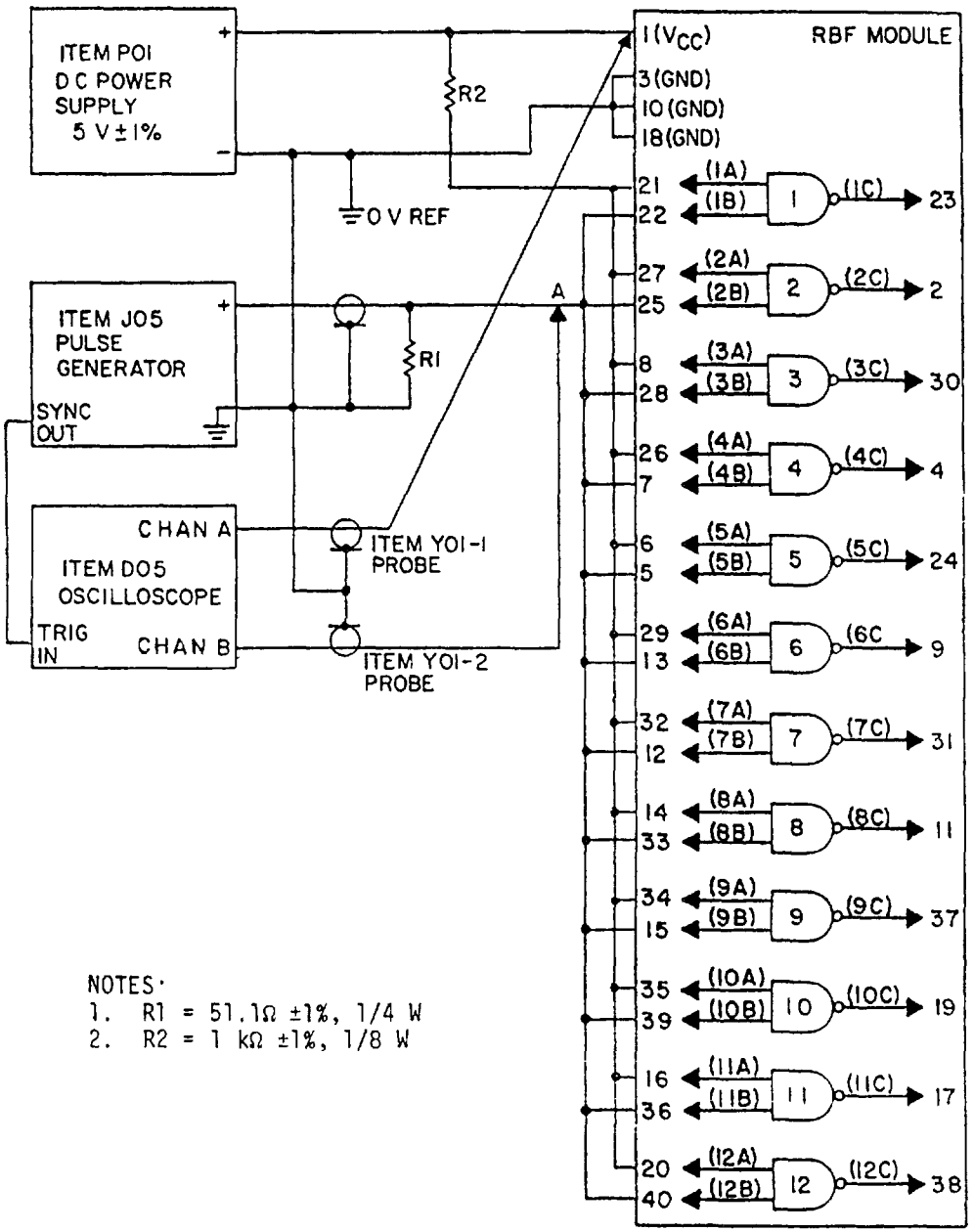
Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
Y1	1A	Input capacitance	21	Open	2	15	pF	25
Y2	1B	"	22	"	"	"	"	"
Y3	2A	"	27	"	"	"	"	"
Y4	2B	"	25	"	"	"	"	"
Y5	3A	"	8	"	"	"	"	"
Y6	3B	"	28	"	"	"	"	"
Y7	4A	"	26	"	"	"	"	"
Y8	4B	"	7	"	"	"	"	"
Y9	5A	"	6	"	"	"	"	"
Y10	5B	"	5	"	"	"	"	"
Y11	6A	"	29	"	"	"	"	"
Y12	6B	"	13	"	"	"	"	"
Y13	7A	"	32	"	"	"	"	"
Y14	7B	"	12	"	"	"	"	"
Y15	8A	"	14	"	"	"	"	"
Y16	8B	"	33	"	"	"	"	"
Y17	9A	"	34	"	"	"	"	"
Y18	9B	"	15	"	"	"	"	"
Y19	10A	"	35	"	"	"	"	"
Y20	10B	"	39	"	"	"	"	"

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TABLE XX. Input capacitance test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
Y21	11A Input capacitance	16	Open	2	15	pF	25
Y22	11B "	36	"	"	"	"	"
Y23	12A "	20	"	"	"	"	"
Y24	12B "	40	"	"	"	"	"

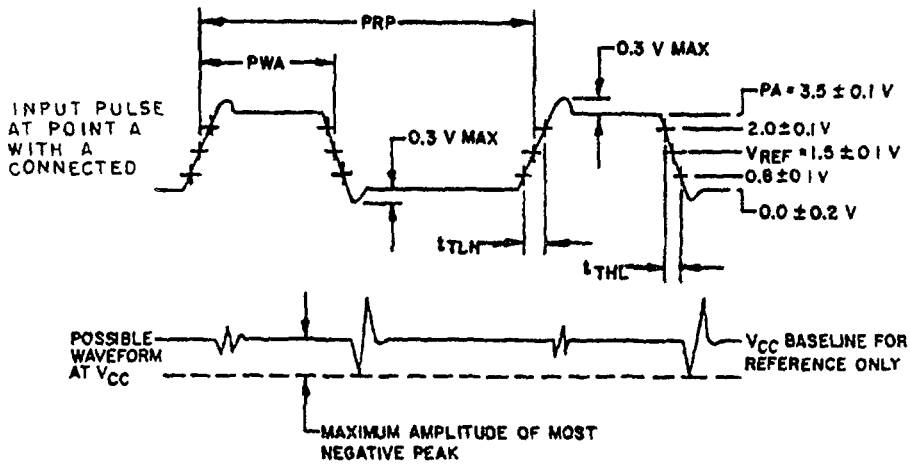
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NOTES:
1. R1 = 51.1Ω ±1%, 1/4 W
2. R2 = 1 kΩ ±1%, 1/8 W

FIGURE 23. Power supply transient amplitude test circuit for table XVIII

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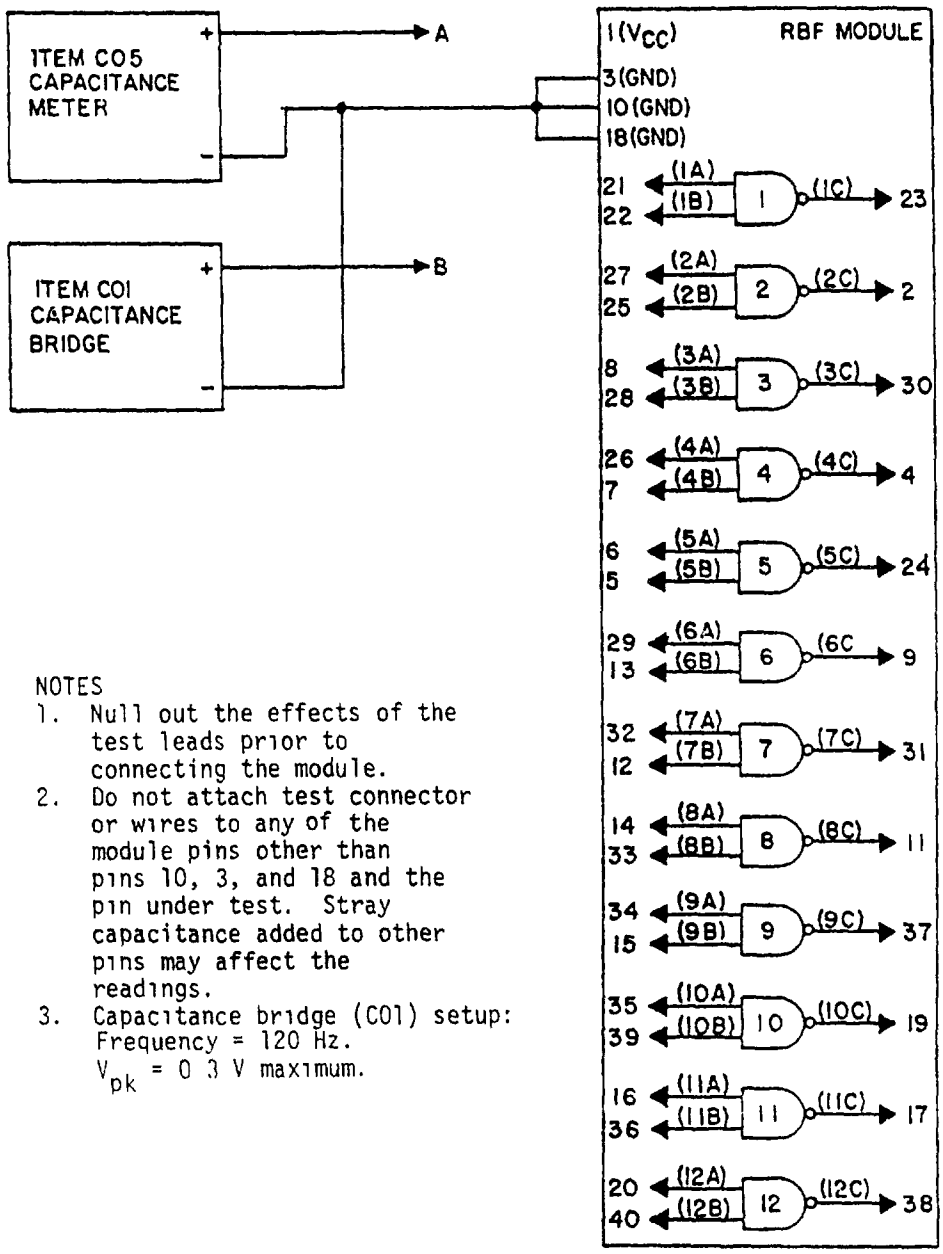


NOTE Input pulse parameters: With the module connected to the pulse generator, adjust the pulse generator for the required pulse amplitude, pulse low level voltage, and transition times. Verify the input pulse parameters at each input pin before making the measurement

$t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (measured at the 0.8 V and 2.0 V points of input pulse).
PWA = 500 ±50 ns.
PRP = 1,000 ±100 ns

FIGURE 24. Power supply transient amplitude test circuit waveforms for table XVIII.

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NOTES
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins other than pins 10, 3, and 18 and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Capacitance bridge (COI) setup: Frequency = 120 Hz. V_{pk} = 0.3 V maximum.

FIGURE 25. Power supply filter and input capacitance test circuit for table XIX and XX.

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